

SYMBOLIC DYNAMICS.

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1. Introduction. In both its qualitative and quantitative aspects, modern theoretical dynamics is largely concerned with recurrence and transitivity. The qualitative study of dynamics owes its inception to Hill and Poincaré. Its development in the past three decades has been dominated by the work of Birkhoff, to whom belongs the principal credit for the trend of modern research in dynamics. The quantitative (metrical) study of dynamical systems is of more recent origin, but with the establishment of the ergodic theorems of Birkhoff and von Neumann, its development has been rapid.

The methods used in the study of recurrence and transitivity frequently combine classical differential analysis with a more abstract symbolic analysis. This involves a characterization of the ordinary dynamical trajectory by an unending sequence of symbols termed a *symbolic trajectory* such that the properties of recurrence and transitivity of the dynamical trajectory are reflected in analogous properties of its symbolic trajectory. For general research of this character one may refer to papers by Hadamard, Birkhoff and the authors. Papers which are special in that they refer to spaces of constant curvature, but which are significant by virtue of the symbolisms involved have been written by Artin, Koebe, Martin, Myrberg and Nielsen. Martin and Robbins have contributed papers on purely symbolic aspects.

In the general theory as developed up to the present, the dividing line between symbolic and differential methods has not been sharply defined. It is a first aim of the present paper to separate the symbolic aspects from the differential aspects and to develop the symbolic theory on its own account. In this way many interesting and important dynamical questions can be given an algebraic formulation quite apart from the classical theories of differential equations.

A symbolic trajectory T is formed from symbols taken from a finite set of generating symbols subject to rules of admissibility which are dependent on the underlying Poincaré fundamental group. The trajectory T taken with one of its symbols is called a *symbolic element*. Symbolic elements are analogues of line elements on an ordinary trajectory. A metric is assigned to the space of symbolic elements and this space turns out to be perfect, compact, and totally disconnected. The necessary first theorems on symbolic recurrence and transitivity are then rapidly developed in a way suggestive of the differential treatment of Birkhoff.

We come to the more novel developments of the paper. Recall that the recurrency index $R(n)$ of T is a minimum number m such that each block of m successive symbols of T contains every block of n successive symbols of T . If $R(n)$ exists for each n , T is termed recurrent (cf. Morse [2], p. 94). In § 7 we are concerned with conditions on $R(n)$. A typical condition is that $R(n) > 2n$ for every non-periodic recurrent T . In § 8 we make a special study of the Morse recurrent trajectory (Morse [2], p. 95) obtaining $R(n)$ explicitly. This is the first and only known evaluation of $R(n)$ for a non-periodic recurrent trajectory. We give a definition of almost periodicity of symbolic trajectories somewhat similar to the ordinary definition of Besicovitch ([1], p. 77) and show that the Morse trajectory is not almost periodic. Most of the known recurrent trajectories are almost periodic. In § 9 we give general methods for deriving new recurrent trajectories from a given recurrent trajectory. For the Morse trajectory, $R(n)/n$ is shown to have an inferior limit of 5.5 and a superior limit of 10. In § 10 we define a recurrent trajectory for which the inferior limit of $R(n)/n$ is finite while the superior limit is $+\infty$.

We continue with a study of transitivity. An admissible symbolic sequence of the form $a_1 a_2 a_3 \cdots$ is termed a *ray*. We assign the Baire metric (cf. Hausdorff [1], p. 101) to the space of rays. Let $\theta(n)$ be the minimum length of a block which contains a copy of each admissible n -block. A ray R which contains a copy of each admissible block is termed transitive. If R is transitive, let $\phi(n)$ be the minimum length of an initial block of R which contains a copy of each admissible n -block. We generalize a theorem of Martin [1] and include a hitherto unpublished theorem of R. Oldenburger on the impossibility that $\phi(n) = \theta(n)$ for two successive values of n ($n > 1$). A typical general theorem states that the set of transitive rays for which

$$\liminf_{n \rightarrow \infty} \frac{\phi(n)}{\theta(n)} = 1$$

is a residual set of the second category.

Symbolic dynamics as the authors conceive it forms one of the three divisions

- (1) representation theory,
- (2) symbolic dynamics,
- (3) existence of space forms,

of the whole theory. The representation theory is concerned with the conditions on space forms under which trajectories admit a one-to-one symbolic representation in terms of which the recurrence or transitivity of the trajectory can be determined. These conditions will involve the Poincaré fundamental

group of the space and differential conditions such as that of uniform instability (cf. Morse [4], p. 64). In (3) one is concerned with the existence of space forms satisfying the conditions discovered in (1). The questions involved are rather deep extensions of the Hilbert, Koebe theory of spaces of negative curvature (cf. Hilbert [1], and Koebe [1]). A simple typical theorem is that there exists no two-dimensional Riemannian manifold of the topological type of the torus satisfying the condition of uniform geodesic instability. The bearing of such studies on questions of topological and metric transitivity will be made clear in later papers.

2. Symbolic sequences. Having seen the significance of symbolic sequences we shall now develop their theory on an independent basis. We start with a finite set of μ distinct symbols termed *generating symbols*. We suppose that $\mu > 1$. If a_n represents a generating symbol, sequences of the form

$$(2.1) \quad \cdots a_{-1} a_0 a_1 \cdots,$$

$$(2.2') \quad a_r a_{r+1} a_{r+2} \cdots,$$

$$(2.2'') \quad \cdots a_{m-2} a_{m-1} a_m,$$

$$(2.3) \quad a_{r+1} a_{r+2} \cdots a_{r+n},$$

will be termed indexed *sequences* (written *I-sequences*). More particularly *I-sequences* of the forms (2.1), (2.2) and (2.3) will be termed *I-trajectories*, *I-rays*, and *I-blocks* respectively. It will be convenient to term the generating symbol represented by a_n the *value* of a_n . The *I-blocks* (2.3) will be said to have the *length* n and will frequently be termed *n-blocks*.

Suppose A and B are *I-sequences* of symbols a_n and b_m respectively. We shall regard A and B as identical and write $A \equiv B$ if and only if the ranges of the indices n and m are the same and a_n and b_n have the same value. Let r be an integer, positive, negative or zero. Suppose that the ranges of n and m are such that symbols a_n and b_m correspond in a one-to-one manner with a_n corresponding to b_{n+r} . If then a_n and b_{n+r} have the same value we shall say that A and B are *similar* and write $A \sim B$. We shall also write

$$B \equiv D_r A, \quad A \equiv D_{-r} B.$$

If $r = 0$, $D_r A \equiv A$, but in general $D_r A$ is different from A when $r \neq 0$.

The class of *I-sequences* similar respectively to an *I-trajectory*, *I-ray*, or *I-block* will be termed a *trajectory*, *ray*, or *block*, and will be represented by the sequence of values involved without indices. For example if 1 and 2 are generating symbols, the unending sequence

$$\cdots 121212 \cdots$$

is a trajectory but not an *I-trajectory*.

Corresponding to an I -sequence A we shall refer to I -subblocks or I -subrays. We shall thereby mean subsets of consecutive symbols of A 'indexed as' in A with successive indices differing by unity. Thus if A is of the form

$$a_1 a_2 a_3 a_4,$$

$a_2 a_3$ is an I -subblock, but $a_1 a_3$ is not an I -subblock.

In classical dynamics an element based on a motion T is a point P of T together with the direction tangent to T at P . The symbolic analogue of such an element is defined as follows. Let (a) be an I -trajectory and a_r the r -th symbol of (a) . The I -trajectory (a) taken with a_r will be termed an I -element based on (a) at a_r , and will be denoted by $E(r, a)$. We term r the *index* of $E(r, a)$ and a_r the *preferred* symbol. Let $E(s, b)$ be an I -element of index s based on an I -trajectory (b) . We shall regard $E(r, a)$ and $E(s, b)$ as identical and write

$$E(r, a) \equiv E(s, b)$$

if and only if $(a) \equiv (b)$ and $r = s$. We shall say that $E(r, a)$ is similar to $E(s, b)$ and write $E(r, a) \sim E(s, b)$ if

$$D_{-r}(a) \equiv D_{-s}(b).$$

The class of I -elements similar to a given I -element $E(r, a)$ will be termed a *symbolic element* E . Two symbolic elements are regarded as identical if the corresponding classes of I -elements are identical. We say that E is based on (a) and represented by $E(r, a)$.

We shall give an illustrative example. Let (a) be an I -trajectory in which a_n equals 1 or 2 according as n is even or odd, and let (b) be an I -trajectory in which $b_{n+1} = a_n$. Evidently

$$E(0, a) \not\equiv E(0, b), \quad E(0, a) \sim E(1, b).$$

More generally if n is even and m is odd,

$$E(n, a) \sim E(m, b)$$

and these similar elements serve to define a symbolic element E . There are just two different symbolic elements based on (a) or (b) .

3. The space of symbolic elements. We shall assign a distance $E_1 E_2$ to any two symbolic elements E_1, E_2 . To that end we say that an I -block of the form

$$c_{r-m} \cdots c_r \cdots c_{r+m}$$

is centered at c_r . Let E_1 and E_2 be represented by indexed elements $E(r, a)$ and $E(s, b)$ respectively. If $E_1 \not\equiv E_2$, let m be the length of the longest I -blocks of (a) and (b) which are similar and centered at a_r and b_s respectively. We then set

$$(3.1) \quad E_1 E_2 = E(r, a) E(s, b) = 1/m \quad (m \neq 0).$$

In the special case where $m = 0$ we regard the distance (3.1) as infinite. If $E_1 \equiv E_2$, the distance (3.1) is taken as zero. It is clear that the distance $E_1 E_2$ is independent of the particular I -elements $E(r, a)$ and $E(s, b)$ chosen to represent E_1 and E_2 respectively.

The distances so defined satisfy the usual metric axioms:

$$(3.2) \quad E_1 E_2 = 0 \text{ if and only if } E_1 \equiv E_2,$$

$$(3.3) \quad E_1 E_2 = E_2 E_1,$$

$$(3.4) \quad E_1 E_3 \leq E_1 E_2 + E_2 E_3.$$

Relation (3.4) follows from the stronger relation

$$(3.5) \quad E_1 E_3 \leq \max(E_1 E_2, E_2 E_3).$$

In case any one of the distances involved in (3.5) is 0 or ∞ , (3.5) holds trivially as is readily seen. Excepting these cases, let h , k , and m be the reciprocals of $E_1 E_3$, $E_1 E_2$ and $E_2 E_3$ respectively. Consider the case $k \geq m$. Evidently then $h \geq m$ so that $h \geq \min(k, m)$. The latter relation yields the condition

$$1/h \leq \max(1/k, 1/m)$$

from which (3.5) follows. The case $m \geq k$ leads to the same result.

Recall that a metric space M is compact if each infinite sequence of points of M contains at least one subsequence which converges to a point of M . With this understood we shall prove the following.

THEOREM 3.1. *The space M of all symbolic elements is compact.*

Let E_n be the n -th element in an arbitrary sequence of symbolic elements. Suppose E_n is based on the I -trajectory (a^n) . Without loss of generality we can suppose that the preferred symbol in (a^n) has the index 0. We shall define a doubly infinite matrix of I -trajectories

$$(3.6) \quad T_{ij} \quad (i, j = 1, 2, \dots)$$

as follows. Since there is at most a finite number of generating symbols there exists an infinite subset of the symbols

$$a_0^n \quad (n = 1, 2, \dots)$$

which have the same "value" c_0 . Let the I -trajectories T_{1j} be a subsequence of the I -trajectories (a^n) in which a_0^n has the value c_0 . Proceeding inductively we suppose that the first $k - 1$ rows of the matrix (3. 6) have been determined. The k -th row shall then be a subsequence T_{ki} of the $(k - 1)$ -st row such that the symbols of T_{ki} of index $k - 1$ have a common value c_{k-1} , while the symbols of index $1 - k$ have a common value c_{1-k} .

Let (c) be the I -trajectory

$$\cdots c_{-1} c_0 c_1 \cdots$$

Then (c) has an I -block of length $2k - 1$, with center at c_0 , in common with T_{kk} . Let $\mathcal{E}_k$ be the element of index 0 based on T_{kk} , and let $\mathcal{E}$ be the element of index 0 based on (c) . Then

$$\mathcal{E}\mathcal{E}_k \leq \frac{1}{2k-1},$$

so that the elements $\mathcal{E}_k$ converge to $\mathcal{E}$. Moreover the elements $\mathcal{E}_k$ are a subsequence of the given elements $\mathcal{E}_n$, and the proof of the theorem is complete.

A metric space is termed *totally disconnected* if its connected subsets all reduce to points. Cf. Hausdorff [1], p. 152. Recall also that if C is a connected set and d is a distance assumed by a pair of points of C , there exists a pair of points of C which assumes an arbitrary distance between 0 and d . In the case of the space M of symbolic elements, the distances take on only the discrete values $1/n$, so that the space M is totally disconnected.

Finally, the space M is perfect. We already know that M is compact. To prove that M is perfect it is necessary to show that in every neighborhood of an indexed element $E(0, a)$ there is an I -element $E(0, b)$ not similar to $E(0, a)$. To that end we take (b) so that it has a prescribed block A of (a) centered at a_0 , in common with (a) , but that all longer blocks of (a) with centers at a_0 differ from corresponding blocks of (b) . The element $E(0, b)$ will not be similar to $E(0, a)$ but will have an arbitrarily small distance from $E(0, a)$ if the block A is sufficiently long. We accordingly have the following theorem.

THEOREM 3.2. *The space of all symbolic elements is compact, perfect and totally disconnected.*

By virtue of a well known theorem in point set theory (cf. Hausdorff [1], p. 160), Theorem 3.2 has the following corollary.

COROLLARY. *The space of all symbolic elements is the homeomorph of the Cantor, perfect, nowhere dense, linear set.*

4. The space of rays and the space of I -trajectories. Let R_1 and R_2 be rays with indexed representations of the form

$$a_r a_{r+1} \cdot \cdot \cdot .$$

If $R_1 \not\equiv R_2$, the rays R_1 and R_2 have common initial blocks of finite maximum length m . With Baire (cf. Hausdorff, p. 101) we then set

$$R_1 R_2 = \frac{1}{m+1}.$$

If $R_1 \equiv R_2$, we set $R_1 R_2 = 0$. The distance so defined satisfies the usual metric axioms including a stronger triangle relation similar to (3.5). One readily sees that the space of rays is compact, perfect and totally disconnected.

This distance function in the space of rays is similar to that previously defined in the space of elements. We define a distance between I -trajectories which is based on an entirely different concept and is analogous to the Besicovitch metric of the theory of almost periodic functions (cf. Besicovitch [1], p. 73).

Let (a) and (b) be I -trajectories. Let $|a_i - b_i|$ be 0 if a_i and b_i have the same symbolic value, and 1 if this is not the case. We then set

$$[a, b] = \lim_{n \rightarrow \infty} \sup \frac{1}{2n+1} \sum_{-n}^n |a_i - b_i|.$$

It is easily shown that this distance function satisfies the usual metric axioms with the exception that $[a, b] = 0$ does not imply $(a) \equiv (b)$. A simple example proving this last statement is obtained by letting all the symbols of (a) be 1, and all those of (b) be 1 with the exception of the symbol of index 0 which shall be 2.

If $(a') \sim (a)$ and $(b') \sim (b)$, it will not in general be true that $[a', b'] = [a, b]$, but corresponding to (a') there always exists a $(b') \sim (b)$ such that the equality does hold. Recalling the notation of § 2, let $D_r(a)$ denote the I -trajectory (a') in which $a'_{i+r} = a_i$. Thus $D_r(a) \sim (a)$. Furthermore, if $(a') \sim (a)$, there exists an r such that $D_r(a) \equiv (a')$. We shall establish the following lemma.

LEMMA 4.1. *If (a) and (b) are two I -trajectories and r is integral,*

$$(4.1) \quad [D_r(a), D_r(b)] = [a, b].$$

Since $D_r(D_{-r}(a)) = (a)$, it is sufficient to prove (4.1) under the assumption that $r > 0$. Denoting $D_r(b)$ by (b') we have

$$\begin{aligned} \frac{1}{2n+1} \sum_{-n}^n |a'_i - b'_i| &= \frac{1}{2n+1} \sum_{-n-r}^{n-r} |a_i - b_i| \\ &= \frac{1}{2n+1} \sum_{-n}^{+n} |a_i - b_i| + \frac{1}{2n+1} \sum_{-n-r}^{-n-1} |a_i - b_i| - \frac{1}{2n+1} \sum_{n-r+1}^n |a_i - b_i|. \end{aligned}$$

Since $0 \leq |a_i - b_i| \leq 1$, the last two terms approach 0 as n becomes infinite. Thus

$$\lim_{n \rightarrow \infty} \sup. \frac{1}{2n+1} \sum_{-n}^n |a'_i - b'_i| = \lim_{n \rightarrow \infty} \sup. \frac{1}{2n+1} \sum_{-n}^n |a_i - b_i|,$$

which is the stated result.

Let T be a trajectory and let (a) be an I -trajectory representing T . Given $\epsilon > 0$, T will be said to admit the ϵ -translation r if $[D_r(a), a] < \epsilon$. From (4.1), this definition is independent of the I -trajectory chosen to represent T . Any trajectory admits the ϵ -translation 0, and this will be termed the *trivial* ϵ -translation.

As in the theory of almost periodic functions, a set of integers $\dots < r_{-1} < r_0 < r_1 < r_2 < \dots$ is termed *relatively dense* if the numbers $|r_i - r_{i-1}|$, $i = 0, \pm 1, \dots$, form a bounded set. We define the trajectory T as *almost periodic* if given $\epsilon > 0$, T admits a relatively dense set of ϵ -translations. A periodic trajectory is evidently almost periodic. We shall presently define recurrent trajectories and shall use the notion of almost periodicity in the classification of such trajectories. Most of the known examples of recurrent trajectories turn out to be almost periodic, but the Morse recurrent trajectory of § 8 is not almost periodic, as we shall see.

5. Restricted sequences. In the applications to dynamical theory the sequences admitted are not formed freely from the generating symbols, but are subject to certain limitations. The simplest of these restrictions may be called the *restriction of inverses*. In this case the generating symbols form a set of the form

$$a_1, \dots, a_p, a_1^{-1}, \dots, a_p^{-1}, \quad (p > 1)$$

in which a_i^{-1} is termed the inverse of a_i . In this case a sequence is inadmissible if a symbol and its successor are inverses. In other cases a circular order C of the generating symbols may appear and an integer r , with the restriction that p -blocks in the order C with $p > r$ are to be excluded. Another type of restriction prohibits successive blocks of certain definite forms.

In general problems restrictions of all of these types will appear. What these restrictions are in any problem depends upon the underlying group theory and topology. A detailed study of these restrictions will appear in a later

paper. In the present paper we are fortunately able to abstract the essentials of these conditions as follows.

Certain blocks of the generating symbols will be distinguished and termed admissible. A ray or trajectory all of whose subblocks are admissible will be termed admissible as well as any element based on an admissible trajectory. The conditions of admissibility on blocks are as follows.

(α). A block is admissible if and only if all its subblocks are admissible.

(β). Corresponding to any admissible block A there exist at least two generating symbols e , dependent at most upon the last two symbols of A and such that Ae is admissible. A similar condition holds replacing last by first and Ae by eA .

(γ). Corresponding to any two admissible blocks A and B there exists at least one generating symbol e dependent at most upon the last two symbols of A and the first two symbols of B and such that AeB is admissible.

(δ). There exist at least two generating symbols α and β such that all blocks formed from α and β alone are admissible.

These conditions are satisfied when there are at least two generating symbols and their inverses and the only restriction is the restriction of inverses. They are also satisfied by the well known Nielsen symbolism (cf. Nielsen [2], pp. 211-217) for surfaces of genus $p > 1$. Various symbolisms developed by the authors satisfy these conditions, including one which replaces the Nielsen symbolism to advantage.

We state the following theorem.

THEOREM 5.1. *The set of admissible elements form a compact, perfect, totally disconnected subset Ω of the space M of all elements.*

That Ω is compact may be seen as follows. Let E_n be a sequence of admissible elements based respectively on trajectories t_n . Let E be a limit element of the sequence E_n . Suppose that E is based on the trajectory t . Each block in t must appear in some trajectory t_n and accordingly be admissible. Hence t and E are admissible, and Ω is compact.

To show that Ω is perfect let E be an arbitrary admissible element. Represent E by an I -element $E(0, a)$. Let A be a $(2n+1)$ -block of (a) centered at a_0 . By virtue of admissibility condition (β), the block A can be extended to form an admissible I -trajectory of the form

$$(5.1) \quad \cdots e_{-n-2}e_{-n-1}Ae_{n+1}e_{n+2} \cdots,$$

where the symbols e_i and e_{-i} , $i > n$, can be successively chosen in two different

ways. Let E' be an element of index 0 based on (5.1). With the freedom of choice of the symbols e_i and e_{-i} it appears that the elements E' include elements different from E . But these elements also include elements arbitrarily near E since A is arbitrarily long.

The set Ω is accordingly perfect. It is totally disconnected since the space M is totally disconnected, and the proof of the theorem is complete.

A theorem similar to the above is true of the set of admissible rays, taking the metric of rays into account. Unless otherwise stated, the only trajectories and rays which will be used will be those which are admissible in the sense of this section.

6. Limit trajectories and minimal trajectories. Let t be a trajectory represented by an I -trajectory (a) . If there exists a positive integer r such that

$$a_{n+r} = a_n \quad (n = 0, \pm 1, \pm 2, \dots),$$

t will be said to have the *period* r . A trajectory of period r is determined by each of its r -blocks. There are accordingly at most countably infinite many distinct periodic trajectories.

Let Σ be an arbitrary set of admissible trajectories. A trajectory t will be termed a *limit trajectory* of Σ if the elements based on trajectories of Σ include among their limit elements at least one element based on t . The set Σ can consist of a single trajectory, for example of the trajectory t given by

$$(6.1) \quad \dots 2220220220220222 \dots$$

The trajectory $\dots 222 \dots$ is a limit trajectory of the trajectory (6.1).

A periodic trajectory has no limit trajectory because it bears at most a finite number of elements. Let t be a non-periodic trajectory, and (a) an I -trajectory representing t . Any two I -elements with indices n and r respectively based on (a) , with $n > r$, are dissimilar. For a similarity relation between these two elements would imply that (a) had the period $n - r$, contrary to hypothesis. The elements based on t with positive indices must then have at least one limit element, and we term such elements *positive limit elements* of t . *Negative limit elements* are similarly defined.

A trajectory can have more than one distinct limit trajectory. Letting a^n stand for a block $a \dots a$ of length n we see that a trajectory of the form

$$(6.2) \quad \dots \beta^4 \alpha^3 \beta^2 \alpha \beta^2 \alpha^3 \beta^4 \dots$$

has the trajectories $\dots \alpha \alpha \alpha \dots$ and $\dots \beta \beta \beta \dots$ as limit trajectories. More generally let the set of admissible blocks be enumerated in the form

$$A_0, A_1, \dots$$

If generating symbols e_n and b_n are successively chosen in accordance with block condition (γ) the trajectory

$$(6.3) \quad \cdots A_2 b_2 A_1 b_1 A_0 e_1 A_1 e_2 A_2 \cdots$$

will be admissible. It will have every admissible trajectory as a limit trajectory. The elements based on the trajectory (6.3) are everywhere dense in the set Ω of admissible elements. Such a trajectory is termed *transitive*.

In the remainder of this section we shall be concerned with a fundamental type of limit trajectories termed *minimal* trajectories. The dynamical counterpart of such trajectories was first introduced by Birkhoff [1]. Birkhoff also developed their general theory. In this connection the symbolic method was first used by Morse ([1], [2]) to establish the existence of non-periodic recurrent motions. Theorems 6.1 and 6.3 of this section and Theorem 7.2 of the following section are the abstract symbolic analogues of dynamical theorems of Birkhoff. The proofs presented here are independent of the theory of differential equations.

Let s be an admissible trajectory and n an arbitrary positive integer. The number of different n -blocks of s is finite. It will be denoted by $P_s(n)$ and termed the n -th *permutation number* of s . If t is a limit trajectory of s ,

$$(6.4) \quad P_s(n) \geq P_t(n).$$

A trajectory s such that the equality holds in (6.4) for every n and every limit trajectory t of s will be termed *minimal*.

We understand that periodic trajectories are minimal. They satisfy the above definition vacuously since they have no limit trajectories. We continue with a proof of the following theorem.

THEOREM 6.1. *Among the positive (or negative) limit trajectories of an admissible non-periodic trajectory T there is at least one admissible minimal trajectory.*

We shall give the proof for the case of positive limit trajectories. To that end we shall define certain symbols inductively as follows:

H_0 = the set of positive limit trajectories of T .

p_1 = the minimum of $P_t(1)$ for t in H_0 .

H_1 = the subset of H_0 for which $P_t(1) = p_1$.

$\cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot$

p_n = the minimum of $P_t(n)$ for t in H_{n-1} .

H_n = the subset of H_{n-1} for which $P_t(n) = p_n$.

The set H_0 is not empty since T is non-periodic. Hence H_n is not empty. Moreover

$$H_0 \supset H_1 \supset H_2 \cdots$$

We continue with a proof of the following statement.

(α). *The set H_n is closed in the sense that it includes its limit trajectories.*

That H_0 is closed follows from the definition of limit trajectories. Proceeding inductively we shall assume that H_{n-1} is closed and prove that H_n is closed. In case the set H_n has no limit trajectories, H_n is certainly closed. Suppose then that H_n has a limit trajectory t . To show that t is in H_n , recall that $H_{n-1} \supset H_n$ and that t is accordingly a limit trajectory of H_{n-1} . Since H_{n-1} is assumed to be closed t is in H_{n-1} . By virtue of the definition of p_n ,

$$(6.5) \quad P_t(n) \geq p_n.$$

On the other hand, t is a limit trajectory of trajectories of H_n , and the n -th permutation numbers of trajectories of H_n equal p_n . Hence

$$(6.6) \quad P_t(n) \leq p_n.$$

Hence the equality prevails in (6.5) and (6.6), and t is in H_n . Thus H_n is closed.

The sets H_n form a decreasing sequence of closed sets. It follows from the compactness of the space of admissible elements that the intersection H of the sets H_n is closed and non-void. Let s be a trajectory of H . If s is periodic, s is minimal and the proof is complete. If s is not periodic, let t be a limit trajectory of s . Since t and s are both in H_n for every n , t and s have the same n -th permutation number for every n . Hence s is minimal, and the proof is complete.

THEOREM 6.2. *If s is a minimal trajectory and t a limit trajectory of s , then t is a minimal trajectory and s a limit trajectory of t .*

Every n -block of s is in t since s is minimal, so that s is a limit trajectory of t . To show that t is minimal, let r be one of its limit trajectories. Then r is also a limit trajectory of s . Since s is minimal, the n -th permutation numbers of t and r equal that of s , and so equal each other. Hence t is minimal, and the proof is complete.

The set of limit trajectories of a trajectory t will be termed the *derived set* of t . If s is a non-periodic minimal trajectory, each member of its derived set S has s as a limit trajectory, and hence has the same derived set. That

this property is characteristic of non-periodic minimal trajectories is implied by the following theorem.

THEOREM 6.3. *A set S of trajectories which is the derived set of each of its members consists of non-periodic minimal trajectories.*

Since the derived set of a periodic trajectory is empty, no trajectory of the set S can be periodic.

If s is a trajectory of S and t is a limit trajectory of s ,

$$(6.7) \quad P_s(n) \geq P_t(n).$$

But t is in S and thus s is a limit trajectory of t , so that the equality sign must hold in (6.7). Hence s is minimal and the proof is complete.

The elements based on trajectories of S form a compact, perfect, totally disconnected set H . This is behind an earlier result of Morse that there exist discontinuous recurrent motions. Cf. Morse [2], p. 99. The set H is the homeomorph of the Cantor, perfect, nowhere dense, linear set. In particular H has the power $\aleph$ of the continuum. Since there is at most a countable infinity $\aleph_0$ of elements based on each trajectory of S we can conclude that the trajectories of S have a power a such that $\aleph_0 a = \aleph$. But $\aleph_0 a = a$ (cf. Hausdorff, pp. 30-31) so that $a = \aleph$. The set S thus has the power of the continuum.

We shall refer to the following definition. A set of trajectories which is the derived set of each of its members or which reduces to a single periodic trajectory is termed a *minimal set* of trajectories.

7. Recurrent trajectories. Let h be an admissible sequence, finite or infinite, and n a positive integer. A least integer m (if any exists) such that each m -block of h contains every n -block of h will be called the n -th *recurrency index* $R_h(n)$ of h . If h is a ray or a trajectory and $R_h(n)$ exists for each n , h will be termed *recurrent* and $R_h(n)$ the *recurrency function* of h .

We begin with the following theorem.

THEOREM 7.1. *If s is a recurrent trajectory and t is a limit trajectory of s , then t is recurrent and s is a limit trajectory of t . Moreover*

$$R_s(n) = R_t(n).$$

Each block of t appears in s since t is a limit trajectory of s . Let A be an arbitrary n -block of s . A copy of A is found in each block of s of length $R_s(n)$, and hence in each block of t of length $R_s(n)$. Hence $R_t(n)$ exists with

$$(7.1) \quad R_t(n) \leq R_s(n).$$

Since each block A of s appears in t , s is a limit trajectory of t . The rôles of s and t can then be interchanged so that the equality prevails in (7.1). The proof of the theorem is complete.

The following theorem is the symbolic analogue of a theorem of Birkhoff. Cf. Birkhoff [2], p. 199.

THEOREM 7.2. *A necessary and sufficient condition that a trajectory s be recurrent is that it be minimal.*

We begin by assuming that s is minimal but not recurrent, and seek a contradiction. If s is not recurrent, there exists a block U of s with the following property. There exists a sequence A_n of blocks of s whose lengths become infinite with n and which do not contain U . Without loss of generality we can suppose that the length of A_n is odd. Let E_n then be the element based on s with preferred symbol at the center of A_n . Since the space of admissible elements is compact there exists a subsequence of the elements E_n which converges to an admissible element E . Let t be the trajectory on which E is based. It is clear that t does not contain the block U . Hence $t \neq s$. It follows that t is a limit trajectory of s . Since s is minimal, t and s must then contain the same blocks. From this contradiction we infer that s is recurrent.

It remains to show that s is minimal if recurrent. If s is periodic, s is minimal. Assume then that s is not periodic and let S be the derived set of s . It follows from Theorem 7.1 that S is the derived set of each of its members. We conclude from Theorem 6.3 that s is minimal, and the proof is complete.

It is natural to ask what limitations on recurrency indices $R(n)$ and permutation indices $P(n)$ are imposed by conditions of periodicity or recurrency. We give a partial answer in the following lemmas and theorems.

LEMMA 7.1. *If $P(n)$ is the number of different n -blocks in a trajectory t , $P(n+1) \geq P(n)$.*

Let the different n -blocks in t be enumerated in the form

$$(7.2) \quad B_i \quad [i = 1, 2, \dots, P(n)],$$

and let e_i be the successor of B_i in some position in t . The $(n+1)$ -blocks $B_i e_i$ are all different, and the lemma follows immediately.

If (a) is an indexed trajectory, a block B of (a) whose first and last symbols are a_r and a_m will be denoted by $[r, m]$. We shall term r the *first index* of B and m the *final index* of B .

THEOREM 7.3. *A trajectory t for which $P(m+1) = P(m)$ for some m has a period $\omega \leq P(m)$.*

Let (a) be an indexed representation of t . There are $P(m)$ dissimilar m -blocks B_i in (a) . The elements immediately following copies of B_i in (a) must always be the same. Otherwise there would be at least $P(m) + 1$ dissimilar $(m + 1)$ -blocks. Similarly the elements immediately preceding copies of B_i must always be the same.

Consider the blocks $B = [i + 1, i + m]$. Since there are just $P(m)$ dissimilar m -blocks, for some i there must exist a similarity relation of the form

$$(7.3) \quad [i + 1, i + m] \sim [i + 1 + \omega, i + m + \omega],$$

in which $0 < \omega \leq P(m)$. We shall show that (a) has a period ω . We begin with an inductive proof of the relations,

$$(7.4) \quad a_{i+r} = a_{i+r+\omega} \quad (r = 1, 2, \dots).$$

That (7.4) holds for $r = 1, 2, \dots, m$ follows from (7.3). Assuming that (7.4) holds for $r = 1, 2, \dots, n \leq m$, we shall prove that (7.4) holds for $r = n + 1$. The m -blocks with final indices $i + n$ and $i + n + \omega$ respectively are similar by virtue of our inductive hypothesis. It then follows from the principle stated in the first paragraph of this proof that (7.4) holds for $r = n + 1$.

It follows similarly that $a_{i+r} = a_{i+r+\omega}$ for $r \leq 0$. We thereby make use of the fact that two similar m -blocks must be preceded by symbols with equal values. One starts with the similarity relation (7.3) and proceeds inductively. We conclude that (a) has the period ω , and the proof is complete.

COROLLARY. For a non-periodic trajectory $P(m + 1) > P(m)$ for each $m > 0$.

There exist trajectories for which $P(n + 1) = P(n) + 1$, so that the relation $P(n + 1) > P(n)$ of the corollary is in a sense the best possible. An example of a recurrent trajectory for which $P(n + 1) = P(n) + 1$ for each $n > 0$ will be given in a later paper.

LEMMA 7.2. If t is a trajectory which involves μ generating symbols and has no period less than ω , then

$$(7.5) \quad P(n) \geq n + \mu - 1$$

for all values n for which $P(n) < \omega$.

Relation (7.5) is clearly true when $n = 1$. From the preceding theorem $P(m + 1) > P(m)$ for each m for which $P(m) < \omega$. Relation (7.5) follows inductively.

THEOREM 7.4. *If t is a non-periodic trajectory involving μ generating symbols,*

$$P(n) \geq n + \mu - 1.$$

LEMMA 7.3. *If t is a recurrent trajectory with no period less than ω , then*

$$(7.6) \quad R(n) \geq P(n) + n \geq 2n + \mu - 1$$

for each n such that $P(n) < \omega$.

Suppose that $P(n) < \omega$ and set $R(n) = m$. Since t is recurrent, each m -block B contains each of the $P(n)$ n -blocks of t . Hence

$$(7.7) \quad m \geq P(n) + n - 1.$$

We shall exclude the equality in (7.7).

Suppose B has the form $[r+1, r+m]$ in an indexed representation (a) of t . Suppose $m = P(n) + n - 1$. Then no two n -blocks of B are similar. In particular the initial n -block A of B does not appear twice in B . But $R(n) = m$ so that A appears in the I -block $[r+2, r+m+1]$, and since A appears but once in B , A appears with final index $r+m+1$. Since A appears with final index $r+n$ as well, $a_{r+n} = a_{r+m+1}$ for arbitrary r . Hence t has a period

$$m - n + 1 = [P(n) + n - 1] - n + 1 = P(n) < \omega.$$

This is contrary to the hypothesis that t has no period less than ω . Hence $m > P(n) + n - 1$, and (7.6) follows with the aid of (7.5).

Lemma 7.3 and Theorem 7.4 yield the following theorem.

THEOREM 7.5. *If t is a non-periodic recurrent trajectory involving μ generating symbols,*

$$(7.8) \quad R(n) \geq P(n) + n \geq 2n + \mu - 1,$$

where $R(n)$ and $P(n)$ are respectively the recurrency and permutation indices of t .

It will be convenient to set $P(0) = 0$ for each trajectory t .

THEOREM 7.6. *If a trajectory t involves μ generating symbols and has a minimum period ω ,*

$$(7.9) \quad R(n) \leq n + \omega - 1 \quad (n = 1, 2, \dots).$$

Moreover there exists a largest integer m such that $P(m) < \omega$. For such an m ,

$$(7.10) \quad R(n) \geq 2n + \mu - 1 \quad (n = 1, 2, \dots, m \leq \omega - \mu),$$

$$(7.11) \quad R(n) = n + \omega - 1 \quad (n = m + 1, m + 2, \dots).$$

Relation (7.9) is immediate. To establish the existence of a largest integer m such that $P(m) < \omega$, observe that each integer $n > 0$ for which $P(n) < \omega$ satisfies the relations

$$2n + \mu - 1 \leq R(n) \leq n + \omega - 1,$$

by virtue of (7.6) and (7.9). It follows that m exists and is at most $\omega - \mu$. Relation (7.10) follows from (7.6).

To establish (7.11) let n be any integer $> m$. Then $P(n) \geq \omega$ and

$$(7.12) \quad R(n) \geq P(n) + n - 1 \geq \omega + n - 1.$$

Relation (7.11) follows from (7.12) and (7.9).

Let (a) be an arbitrary I -trajectory. Let n be a positive integer, and let m be an integer, positive, negative or zero. For n and m fixed, let $H(m)$ be the minimum length of those blocks of (a) with initial index m which contain copies of all n -blocks of (a) . If no such block with initial index m exists, let $H(m) = +\infty$. If, as m ranges over all integers, the set $H(m)$ is bounded, it is clear that the n -th recurrency index $R(n)$ of (a) exists and

$$(7.13) \quad R(n) = \max. H(m).$$

Let r be any integer such that $H(r+1)$ assumes the maximum of $H(m)$. The $R(n)$ -block $[r+1, r+R(n)]$ will then be termed a *minimax n -covering* in (a) or in the trajectory t represented by (a) . The length of a minimax n -covering B in t is $R(n)$ and the final n -block in B does not otherwise appear in B . These two properties are characteristic provided $R(n)$ exists.

We are led to the following theorem.

THEOREM 7.7. *If t is a trajectory for which $R(n+1)$ and $R(n)$ exist, $R(n+1) > R(n)$.*

To prove the theorem let B be a minimax n -covering in t . Let A be the final n -block in B and e the symbol following A . The $(n+1)$ -block Ae does not appear in B since A has no second appearance in B . Hence $R(n+1) > R(n)$, as stated.

It may happen that $R(n+1) = R(n) + 1$ for infinitely many values of n as the recurrent trajectory of the following section shows.

8. The Morse recurrent trajectory. In this section we shall recall the definition of the non-periodic I -trajectory T first defined by Morse [2], p. 95, and shall give the explicit values of the corresponding recurrency function, denoting this function by $\rho(n)$.

Recall that the symbols α and β can be combined in any way to form admissible blocks. We here take α and β as 1 and 2 respectively. Set

$$(8.0) \quad \begin{array}{ll} a_0 = 1, & b_0 = 2, \\ a_1 = a_0 b_0, & b_1 = b_0 a_0, \\ \cdot & \cdot \\ a_{n+1} = a_n b_n, & b_{n+1} = b_n a_n, \end{array}$$

understanding that the blocks $a_n b_n$ and $b_n a_n$ are to be expanded as blocks of 1's and 2's of length 2^{n+1} . The I -trajectory T shall have the representation

$$(8.1) \quad \cdot \cdot \cdot c_{-2} c_{-1} c_0 c_1 c_2 \cdot \cdot \cdot$$

In T we set

$$(8.2) \quad c_0 c_1 \cdot \cdot \cdot c_{2^n-1} \equiv a_n.$$

Note that (8.2) is consistent for all n , since a_m is an initial subblock of a_n for $m < n$. The symbols c_i are thereby defined for $i > 0$. To complete the definition of (8.1) we set

$$(8.3) \quad c_{-r} = c_{r-1} \quad (r > 0).$$

The subray

$$(8.4) \quad R = c_0 c_1 c_2 \cdot \cdot \cdot$$

of T thus starts as follows:

$$(8.5) \quad 1221 \ 2112 \ 2112 \ 1221 \ 2112 \ 1221 \ 1221 \ 2112 \cdot \cdot \cdot$$

Corresponding to an arbitrary block A , the block which consists of the symbols of A in inverse order will be denoted by A^{-1} . We note that

$$(8.6') \quad a_n = a_n^{-1}, \quad b_n = b_n^{-1}, \quad (n \text{ even})$$

$$(8.6'') \quad a_n = b_n^{-1}, \quad b_n = a_n^{-1}, \quad (n \text{ odd}).$$

Relations (8.6) are clearly true when $n = 0$, and their truth in general follows inductively from the relations

$$\begin{aligned} a^{-1}_{m+1} &= (a_m b_m)^{-1} = b_m^{-1} a_m^{-1}, \\ b^{-1}_{m+1} &= (b_m a_m)^{-1} = a_m^{-1} b_m^{-1}. \end{aligned}$$

From (8.6) we infer that the 2^n -block of (8.1) whose final index is -1 is $a_n^{-1} = a_n$ when n is even and $a_n^{-1} = b_n$ when n is odd. Thus the 2^{n+1} -block

whose initial index is -2^n is $a_n a_n$ when n is even and $b_n a_n$ when n is odd. Each block of T appears in such a 2^{n+1} -block for n sufficiently large. Since

$$a_{n+3} = a_n b_n b_n a_n b_n a_n b_n,$$

each block of T appears in the subray R of (8.4). The recurrency indices of T (if they exist) are thus completely determined by this subray.

Let A be an arbitrary block of 1's and 2's. The block obtained from A by replacing 1 by 2 and 2 by 1 will be denoted by A' . An easy inductive proof shows that

$$(8.7) \quad a'_n = b_n, \quad b'_n = a_n.$$

The trajectories T' and T_n . Let T' be a trajectory with an indexed representation (e) in which $e_i = c_i$ for $i \geq 0$, and for $i < 0$ equals 1 or 2 according as c_i is 2 or 1 respectively. The 2^n -block of T whose final index is -1 is a_n or b_n according as n is even or odd. The corresponding 2^n -block of T' is then $a'_n = b_n$ or $b'_n = a_n$ according as n is even or odd. We infer that each block of T' has a copy in the subray R of (8.4). The recurrency indices of T' (if they exist) are then completely determined by R . Since this is also true of T we have proved the following lemma.

LEMMA 8.1. *The recurrency indices of T and T' , if they exist, are equal and are uniquely determined by the ray R in (8.4).*

We introduce a trajectory T_n , $n \geq 0$, whose generating symbols are a_n and b_n , and which is obtained from T or T' according as n is even or odd, by replacing 1 by a_n and 2 by b_n . It follows from the relations (8.0), (8.3) and (8.6) that an expansion of the symbols a_n and b_n in T_n into blocks of 1's and 2's will yield T . Unless otherwise stated T_n shall be regarded as a trajectory in the symbols a_n and b_n unexpanded as blocks of 1's and 2's. For example $a_n b_n$ will be regarded as a 2-block of T_n , or, if we choose, as a 2^{n+1} -block of T .

We shall make use of the following properties of the ray R of (8.4).

- (a). *There are no more than two successive 1's or 2's in R .*
- (b). *Each 5-block of R contains the blocks 12 and 21.*
- (c). *Each 5-block of R contains either the block 11 or else the block 22.*

To establish (a) we note that each 3-block of R will be found in one of the expanded 2-blocks of T_1 ; that is, in one of the blocks

$$\begin{aligned} a_1 b_1 &= 1221, & a_1 a_1 &= 1212, \\ b_1 a_1 &= 2112, & b_1 b_1 &= 2121, \end{aligned}$$

and no more than two successive 1's or 2's appear in these blocks.

To verify (b) observe that a 5-block B which did not contain 12 would contain 21 in its first 3-block by virtue of (a). Its final 3-block would then be 111 contrary to (a). We infer that B contains 12, and similarly 21.

To prove (c) we infer from (a) that there can be no more than two consecutive a_i 's or b_i 's in T_1 . In particular R cannot contain the block 12121 since R would then contain the block $a_1a_1a_1$ or $b_1b_1b_1$ according as 12121 possessed an even or odd first index. Similarly R contains no block 21212, and (c) follows directly.

Statements (a), (b) and (c) are true if R is replaced by T or T' since each block in T or T' appears in the subray R .

We continue with the following lemma.

LEMMA 8.2. *The recurrency index $\rho(n)$ of the trajectory T has the values*

$$\rho(1) = 3, \quad \rho(2) = 9, \quad \rho(3) = 11.$$

That $\rho(1) = 3$ follows from (a). We continue by showing that $\rho(2) \leq 9$. To that end consider a 5-block B of T_1 . The block B will contain the blocks

$$(8.8) \quad a_1b_1 = 1221, \quad b_1a_1 = 2112,$$

by virtue of (b) and either

$$(8.9) \quad a_1a_1 = 1212 \quad \text{or} \quad b_1b_1 = 2121,$$

by virtue of (c). The only 2-blocks in T are 12, 21, 11 and 22. If B_0 denotes the block B expanded in terms of 1's and 2's, we see that each 2-block of T appears in B_0 interior to B_0 . Thus the end symbols of B_0 can be deleted and the resulting block B'_0 contains all 2-blocks of T . But any 9-block of T will contain some block of the nature of B'_0 so that $\rho(2) \leq 9$.

That $\rho(2)$ is not less than 9 follows upon considering the 9-block of T with initial index 10. This block has the form

$$12 \ 1221 \ 211,$$

and will contain no block 11, if the final 1 is removed. Thus $\rho(2) = 9$.

To show that $\rho(3) = 11$, recall that there are no more than two successive 1's or 2's in T so that the only 3-blocks are

$$(8.10) \quad 112, \ 122, \ 121, \ 211, \ 221, \ 212.$$

Consider the block B_0 of the preceding proof. It contains the blocks (8.8) and at least one of the blocks (8.9). It follows that B_0 contains each 3-block (8.10). But any 11-block of T will contain a B_0 , so that $\rho(3) \leq 11$.

That $\rho(3) \geq 11$ follows upon noting that the 11-block of T with initial index 3 has the form

$$1\ 2112\ 2112\ 12,$$

and that removal of the final symbol from this block leaves no subblock of the form 212. Hence $\rho(3) = 11$, and the proof of the lemma is complete.

Let T'_1 be the trajectory obtained from T' by grouping the symbols of T' into 2-blocks with even first indices, regarding T'_1 as generated by the blocks a_1, b_1 . It is readily seen that T'_1 is formed from the blocks a_1, b_1 exactly as T is formed from the symbols 1 and 2 respectively. We shall make use of this fact in proving the following lemma.

LEMMA 8.3. *No two r -blocks of $T(T')$ for which $r \geq 4$ are equal unless their first indices are equal mod 2.*

It is sufficient to prove the lemma for the case $r = 4$. To that end observe that each 4-block of $T(T')$ as it stands is in some 6-block A of $T(T')$ with even first index. Let A_1 be the 3-block of $T_1(T'_1)$ determined by A . Since $\rho(3) = 11$, each 3-block of $T_1(T'_1)$ appears in the 11-block of $T_1(T'_1)$ with first index 0. The lemma follows upon verifying its truth for the 22-block of T with first index 0. For each 4-block of T appears in this 22-block with a first index which is unchanged mod 2.

The following lemma is useful.

LEMMA 8.4. *If $\rho(m+1)$ exists and $m \geq 2$,*

$$\rho(2m) \leq 2\rho(m+1) - 1.$$

We need only to show that each block B of T of length $2\rho(m+1) - 1$ contains a copy of each $2m$ -block x of T . We distinguish two cases.

Case 1. The first index of x is even. Here x determines an m -block of T_1 and so appears in each expanded $\rho(m)$ -block of T_1 . Hence x appears in each $[2\rho(m) + 1]$ -block of T . Thus $\rho(2m)$ exists and

$$2\rho(m) + 1 \geq \rho(2m).$$

But

$$\rho(m+1) \geq \rho(m) + 1,$$

in accordance with Theorem 7.7. Hence

$$2\rho(m+1) - 1 \geq 2\rho(m) + 1 \geq \rho(2m).$$

This completes the proof in Case 1.

Case 2. The first index of x is odd. Let c' and c'' be respectively the elements of T which precede and follow the above block B . If B has an odd first index, $c'B$ has an even first index and a length $2\rho(m+1)$ and so contains a copy of each $(m+1)$ -block of T_1 . Hence $c'B$ must contain a copy y of x . But the index of y is odd in accordance with Lemma 8.3, so that y does not include c' . Hence x appears in B . If B has an even first index, Bc'' contains a copy y of x . But the index of c'' is odd, while the final index of x is even. Hence y does not include c'' and lies in B .

The proof of the lemma is complete.

Since $\rho(3)$ exists it follows from the lemma that $\rho(4)$ exists, then $\rho(6)$, $\rho(10)$, etc. But if $\rho(n)$ exists, $\rho(m)$ exists for $m < n$. We conclude that $\rho(n)$ exists for all n .

We come to a principal theorem.

THEOREM 8.1. *For $m \geq 2$ the recurrency function $\rho(m)$ of T satisfies the relations*

$$(8.11) \quad \rho(2m) = 2\rho(m+1) - 1,$$

$$(8.12) \quad \rho(2m+1) = 2\rho(m+1).$$

Proof of (8.11). It will be sufficient to show that the equality prevails in Lemma 8.4. To that end let g be a minimax $(m+1)$ -covering in T_1 as defined in §7. The length of g is $\rho(m+1)$, and the final $(m+1)$ -block of g appears only once in g . Let x be an arbitrary $2m$ -block of T of odd first index, and let y be an $(m+1)$ -block of T_1 which when expanded contains x . When $m \geq 2$, x appears only as an interior block of the expanded y in accordance with Lemma 8.3. Since a 2-block of T of even index is uniquely determined by either of its symbols we see that y is uniquely determined by x . It thus requires copies of all $(m+1)$ -blocks of T_1 to cover all $2m$ -blocks of T of odd index. In particular it will require a copy of the final $(m+1)$ -block of g . As a block of T , g can then be shortened on the right by at most one symbol and, when expanded, still cover all $2m$ -blocks of T . The equality accordingly holds in Lemma 8.4, and the proof of (8.11) is complete.

Proof of (8.12). Recall that $\rho(2m+1)$ exceeds $\rho(2m)$. It follows then from (8.11) that

$$(8.13) \quad \rho(2m+1) \geq 2\rho(m+1).$$

It remains to show that the equality prevails in (8.13). To that end let B be a $2\rho(m+1)$ -block of T , and let c' and c'' be respectively the symbols pre-

ceding and following B . Let x be an arbitrary $(2m+1)$ -block of T . We distinguish between several cases:

Case 1. The index of c' is even, that of x odd.

Case 2. The index of c' is even, that of x even.

Case 3. The index of c' is odd.

Case 1. The block x appears in some expanded $(m+1)$ -block of T_1 . Hence x has a copy y in $c'B$, for the first $2\rho(m+1)$ -block of $c'B$ determines a $\rho(m+1)$ -block of T_1 . But the index of y is odd so that y does not include c' and thus lies in B .

Case 2. Here c'' has an odd index and the last $2\rho(m+1)$ -block of Bc'' determines a $\rho(m+1)$ -block of T_1 . Hence x has a copy y in Bc'' . But the final index of y is even so that y lies in B .

Case 3. In this case B determines a $\rho(m+1)$ -block of T_1 , hence contains a copy of each $(m+1)$ -block of T_1 , and accordingly a copy of x .

Thus a copy of x appears in B in all cases. The equality accordingly prevails in (8.13), and the proof of (8.12) is complete.

With the aid of the preceding theorem we can derive a general formula for $\rho(n)$ when $n \geq 3$. To that end we first show that integers $n \geq 3$ can be represented in the form

$$(8.14) \quad n = 2^r + p, \quad (p = 2, 3, \dots, 2^r + 1),$$

where $r \geq 0$ and r and p are uniquely determined by n . The smallest and largest values of n given by (8.14) for a fixed r are respectively

$$2^r + 2, \quad 2^{r+1} + 1,$$

and when r is replaced by $r+1$ the smallest value of n is

$$2^{r+1} + 2.$$

Thus as r ranges over the values $0, 1, 2, \dots$ and p over the corresponding values listed in (8.14), n takes on each of the integers $3, 4, \dots$ once and only once. We continue with the proof of the following theorem.

THEOREM 8.2. *For values of n of the form (8.14) the recurrency function $\rho(n)$ of T has the value*

$$(8.15) \quad \rho(n) = 10 \cdot 2^r + p - 1.$$

When $r = 0$ in (8.14) $p = 2$ and $n = 3$. We have seen that $\rho(3) = 11$,

so that (8.15) holds when $n = 3$. When $r = 1$, we see that $p = 2$ and 3 while $n = 4$ and 5 respectively in (8.14). Relation (8.15) affirms that $\rho(4) = 21$ and $\rho(5) = 22$, and these values are correct as follows from (8.11) and (8.12) respectively upon setting $m = 2$. We proceed inductively assuming that (8.15) holds for a fixed r and the related values of p , and seek to prove that

$$(8.16) \quad \rho(\mu) = 10 \cdot 2^{r+1} + q - 1$$

where

$$(8.17) \quad \mu = 2^{r+1} + q \quad (q = 2, \dots, 2^{r+1} + 1).$$

We distinguish between the cases where q is even and q is odd.

Case 1. q even. Here q has the form $2s$ and

$$\mu = 2(2^r + s).$$

Upon setting $m = 2^r + s$ in (8.11) we find that

$$(8.18) \quad \rho(\mu) = 2\rho(2^r + s + 1) - 1.$$

Upon referring to (8.17) we see that $s + 1$ ranges over the values

$$(8.19) \quad 2, 3, \dots, 2^r + 1.$$

When $n = 2^r + s + 1$, $p = s + 1$ in (8.14) and the range (8.19) is also the range prescribed in (8.14). We can accordingly use our inductive hypothesis to infer that

$$(8.20) \quad \rho(2^r + s + 1) = 10 \cdot 2^r + p - 1 = 10 \cdot 2^r + s.$$

Upon returning to (8.18) we conclude that

$$\rho(\mu) = 2(10 \cdot 2^r + s) - 1 = 10 \cdot 2^{r+1} + q - 1.$$

Relation (8.16) thus holds when q is even.

Case 2. q odd. Let q and μ be values of q and μ given by (8.17) with q odd. Corresponding to q and μ there exists a pair of even integers q_1, μ_1 again satisfying (8.17) with $\mu_1 = 2m$ and

$$q = q_1 + 1, \quad \mu = 2m + 1.$$

Upon using the result established under Case 1,

$$\rho(\mu_1) = \rho(2m) = 10 \cdot 2^{r+1} + q_1 - 1.$$

But

$$\rho(2m + 1) = \rho(2m) + 1$$

by virtue of (8.11) and (8.12), so that in Case 2

$$\rho(\mu) = 10 \cdot 2^{r+1} + q_1 = 10 \cdot 2^{r+1} + q - 1.$$

Thus (8.16) holds in Case 2, and the proof of the theorem is complete.

This explicit determination of $\rho(n)$ is the first such determination for a non-periodic, recurrent trajectory.

The manner in which $\rho(n)$ varies with n is illustrated by the following table:

$n = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18,$
 $\rho(n) = 3, 9, 11, \underline{21}, \underline{22}, \underline{41}, \underline{42}, \underline{43}, \underline{44}, \underline{81}, \underline{82}, \underline{83}, \underline{84}, \underline{85}, \underline{86}, \underline{87}, \underline{88}, 161,$
 where the groups underlined are related in an obvious manner. For a given r in (8.14) the smallest and largest values of $\rho(n)$ are respectively

$$10 \cdot 2^r + 1, \quad 11 \cdot 2^r.$$

We see that for a fixed r ,

$$10 > \frac{10 \cdot 2^r + 1}{2^r + 2} \geq \frac{\rho(n)}{n} = \frac{10 \cdot 2^r + p - 1}{2^r + p} \geq \frac{11 \cdot 2^r}{2^{r+1} + 1}.$$

We are thus led to the following theorem.

THEOREM 8.3. *If $\rho(n)$ is the recurrency function of T ,*

$$(8.21) \quad \liminf_{n \rightarrow \infty} \frac{\rho(n)}{n} = \frac{11}{2}, \quad \limsup_{n \rightarrow \infty} \frac{\rho(n)}{n} = 10,$$

and 10 and $11/2$ are the limits respectively of $\rho(n)/n$ for sequences of values of n given by

$$n = 2^r + 2, \quad n = 2^{r+1} + 1, \quad (r = 0, 1, \dots).$$

If ϵ is a prescribed positive constant and r is a sufficiently large positive integer, the ratios

$$\frac{\rho(n)}{n} = \frac{10 \cdot 2^r + p - 1}{2^r + p} \quad (p = 2, 3, \dots, 2^r + 1)$$

assume values within ϵ of each value k between the limits (8.21). There accordingly exists a subsequence of values of n for which

$$\lim_{n \rightarrow \infty} \frac{\rho(n)}{n} = k.$$

For all values of n , $\rho(n) < 10n$, while for each positive ϵ and sufficiently large values of n , $2\rho(n) > (11 - \epsilon)n$.

That the trajectory T is not periodic follows from Theorem 7.6.

In § 4 a trajectory has been defined as almost periodic if it admits a

relatively dense set of ϵ -translations for any positive ϵ . We show that a recurrent trajectory is not necessarily almost periodic by proving that the Morse recurrent trajectory admits only the trivial ϵ -translation if $\epsilon < 1/4$. To that end we first prove two lemmas.

LEMMA 8.5. *If $\cdots \delta_1 \delta_0 \delta_1 \cdots$ is a sequence of the numbers 0 or 1, and p and κ are integers, $\kappa > 0$, then*

$$\limsup_{n \rightarrow \infty} \frac{1}{2n+1} \sum_{i=-n}^n \delta_i = \limsup_{n \rightarrow \infty} \frac{1}{2n+1} \sum_{i=-n}^{n+p} \delta_i = \limsup_{m \rightarrow \infty} \frac{1}{2\kappa m+1} \sum_{i=-\kappa m}^{\kappa m} \delta_i.$$

The first equality follows at once from the relationship

$$\lim_{n \rightarrow \infty} \left| \frac{1}{2n+1} \sum_{i=-n}^n \delta_i - \frac{1}{2n+1} \sum_{i=-n}^{n+p} \delta_i \right| \leq \lim_{n \rightarrow \infty} \frac{p}{2n+1} = 0.$$

As to the second, let $n = \kappa m + r$, $0 \leq r < \kappa$. Then

$$\frac{1}{2n+1} \sum_{i=-n}^n \delta_i = \frac{1}{2n+1} \sum_{i=-\kappa m}^{\kappa m} \delta_i + \frac{1}{2n+1} \sum_{i=-\kappa m-r}^{-\kappa m-1} \delta_i + \frac{1}{2n+1} \sum_{i=\kappa m+1}^{\kappa m+r} \delta_i,$$

where the last two sums are not present if $r = 0$. Each of these last two sums approaches 0 as n becomes infinite. It follows that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{2n+1} \sum_{i=-n}^n \delta_i &= \limsup_{n \rightarrow \infty} \frac{1}{2n+1} \sum_{i=-\kappa m}^{\kappa m} \delta_i \\ &= \limsup_{n \rightarrow \infty} \frac{2\kappa m+1}{2n+1} \cdot \frac{1}{2\kappa m+1} \sum_{i=-\kappa m}^{\kappa m} \delta_i = \limsup_{m \rightarrow \infty} \frac{1}{2\kappa m+1} \sum_{i=-\kappa m}^{\kappa m} \delta_i. \end{aligned}$$

The proof of the lemma is complete.

We recall that T_n is the trajectory with generating symbols a_n and b_n obtained from T or T' , according as n is even or odd, by replacing 1 by a_n and 2 by b_n and that an expansion of the symbols a_n and b_n in T_n into blocks of 1's and 2's will yield T .

LEMMA 8.6. *If $r = 2^p q$, $p \geq 0$, and (c) and (d) are I -trajectories representing T and T_p , respectively,*

$$[D_r(c), (c)] = [D_q(d), (d)].$$

It follows from Lemma 4.1 that the value of $[D_r(c), (c)]$ is independent of the particular indexed representation (c) of T which we use. We shall find it convenient to take (c) as the I -trajectory (8.1). Then $D_r(c)$ is an I -trajectory (c') in which $c'_i = c_{i-r}$. Similarly we can suppose that in (d) , d_i in expanded form is given by

$$d_i = c_{i2^p} \cdots c_{(i+1)2^p-1},$$

for (d) is then an indexed representation of T_p . Since $r = 2^p q$, $D_q(d)$ has the form (d') where $d'_i = d_{i-q} = c_{i2^p-r} \cdots c_{(i+1)2^p-r-1}$, in expanded form.

Since d_i and d'_i are either a_p or b_p , it follows from (8.7) that in their expanded form, d_i and d'_i are either identical, or one is obtained from the other by replacing 1 by 2 and 2 by 1. If we set

$$\{d'_i - d_i\} = \sum_{j=i2^p}^{(i+1)2^p-1} |c'_j - c_j|,$$

it follows that

$$\{d'_i - d_i\} = 2^p |d'_i - d_i|.$$

With the aid of Lemma 8.5, we then see that

$$\begin{aligned} [D_r(c), (c)] &= \lim_{n \rightarrow \infty} \sup \frac{1}{2n+1} \sum_{i=-n}^n |c'_i - c_i| \\ &= \lim_{m \rightarrow \infty} \sup \frac{1}{2 \cdot 2^p m + 1} \sum_{i=-m2^p}^{m2^p} |c'_i - c_i| \\ &= \lim_{m \rightarrow \infty} \sup \frac{1}{2^{p+1}m + 1} \sum_{i=-m2^p}^{m2^p+2^p-1} |c'_i - c_i| \\ &= \lim_{m \rightarrow \infty} \sup \frac{1}{2^{p+1}m + 1} \sum_{i=-m}^m \{d'_i - d_i\} \\ &= \lim_{m \rightarrow \infty} \sup \frac{2^p}{2^{p+1}m + 1} \sum_{i=-m}^m |d'_i - d_i| \\ &= \lim_{m \rightarrow \infty} \sup \frac{1}{2m + 1} \sum_{i=-m}^m |d'_i - d_i| = [D_r(d), (d)]. \end{aligned}$$

The proof of the lemma is complete.

We come to the proof of the desired theorem.

THEOREM 8.4. *The Morse recurrent trajectory admits only the trivial ϵ -translation 0 if $\epsilon < 1/4$.*

It is sufficient to show that

$$(8.22) \quad [D_r(c), (c)] \geq \frac{1}{4}, \quad r \neq 0,$$

where (c) is again the indexed representation (8.1) of the Morse recurrent trajectory.

Case 1. r odd. Let

$$\sigma_n = \sum_{i=n}^{n+3} |c'_i - c_i| = \sum_{i=n}^{n+3} |c_{i-r} - c_i|.$$

Since r is odd, the first indices of $c_{n-r}c_{n-r+1}c_{n-r+2}c_{n-r+3}$ and $c_nc_{n+1}c_{n+2}c_{n+3}$ are not

congruent mod 2. It follows from Lemma 8.3 that these two 4-blocks cannot be similar, so that $\sigma_n \geq 1$. Applying Lemma 8.5, we find that

$$\begin{aligned} [D_r(c), (c)] &= \limsup_{n \rightarrow \infty} \frac{1}{2n+1} \sum_{i=-n}^n |c'_i - c_i| \\ &= \limsup_{n \rightarrow \infty} \frac{1}{8n+1} \sum_{i=-4n}^{4n+3} |c'_i - c_i| \\ &= \limsup_{n \rightarrow \infty} \frac{1}{8n+1} \sum_{i=-n}^n \sigma_{4i}. \end{aligned}$$

Since $\sigma_n \geq 1$, it follows that $\sum_{i=-n}^n \sigma_{4i} \geq 2n+1$, and hence

$$(8.23) \quad [D_r(c), (c)] = \limsup_{n \rightarrow \infty} \frac{1}{8n+1} \sum_{i=-n}^n \sigma_{4i} \geq \lim_{n \rightarrow \infty} \frac{2n+1}{8n+1} = \frac{1}{4}.$$

We have proved (8.22) when r is odd.

Let $(\bar{c})$ be the I -trajectory obtained from (c) by replacing c_n , $n < 0$, by 1 if c_n is 2, and by 2 if c_n is 1. Then $(\bar{c})$ is an indexed representation of the trajectory T' . Since Lemma 8.3 holds for T' , as well as T , a proof similar to that given for (c) yields the relation

$$(8.24) \quad [D_r(\bar{c}), (\bar{c})] \geq \frac{1}{4}, \quad r \text{ odd}.$$

Case 2. r even. Since $r \neq 0$, $r = 2^p q$, where q is odd. From Lemma 8.6,

$$(8.25) \quad [D_r(c), (c)] = [D_q(d), (d)].$$

The symbols of (d) are a_p and b_p , and if in (d) we set $a_p = 1$, $b_p = 2$, we obtain (c) or $(\bar{c})$ according as p is even or odd. Since q is odd, (8.23) or (8.24) applied to the right side of (8.25) yields

$$[D_q(d), (d)] \geq \frac{1}{4}.$$

Combining this inequality with (8.25) we conclude that

$$[D_r(c), (c)] \geq \frac{1}{4}, \quad r \text{ even}.$$

The proof of the theorem is complete.

The following corollary is an immediate consequence of the preceding theorem.

COROLLARY. *The Morse recurrent trajectory is not almost periodic.*

Upon distinguishing recurrent trajectories according to whether they are almost periodic or not, we can state that recurrent trajectories of the type of

Morse are the only ones so far defined which are not almost periodic. In particular, the recurrent trajectories defined by Birkhoff ([4], p. 22) are almost periodic. The proof of this and the further development of the theory of almost periodic trajectories will be given in a later paper by the authors.

9. The derivation of recurrent trajectories. Having exhibited a non-periodic recurrent trajectory T , we shall describe four methods of deriving a recurrent trajectory from T , termed respectively derivation by projection, reduction, association and substitution. We also introduce a general method of derivation which includes each of the preceding. If the given trajectory consists exclusively of the free symbols α and β , these derived trajectories will all be admissible. In general they may or may not be admissible. If the given trajectory T is recurrent, the derived trajectories will be recurrent although usually differing in high degree from T . We suppose that T has an indexed representation (c) in terms of generating symbols $a_1, \dots, a_\mu$.

Projection. The trajectory obtained from (c) by omitting a_k whenever it occurs in (c) will be said to be the k -th projection of (c) .

Reduction. If the generating symbols consist of a finite set of integers, the trajectory obtained by reducing the elements $c_j \bmod p$ will be said to be derived by reduction mod p .

Association. Let m be a positive integer at least 2, and let B_i be the m -block of (c) whose first index is i . There is at most a finite set of different blocks B_i . We introduce the I -trajectory

$$(B) \quad \dots B_{-1} B_0 B_1 \dots,$$

regarding the blocks B_i as the generating symbols.

Substitution. Let $A_1, \dots, A_\mu$ be s -blocks of the generating symbols, $s > 0$. The trajectory (c^*) of a_i 's obtained by replacing a_i in (c) by A_i and expanding will be said to be derived from (c) by block substitution.

Generalized substitution. Let $r_1, \dots, r_m$ be a fixed set of integers, positive, negative, or zero. Let $\phi(x_1, \dots, x_m)$ be a single-valued function (either numerical or symbolic) of its arguments x_i as each x_i ranges over the set $(a_1, \dots, a_\mu)$ of generators of (c) . We include the convention that a null set may be one of the symbolic values of ϕ . We introduce the symbols

$$E_i = \phi[c_{i+r_1}, c_{i+r_2}, \dots, c_{i+r_m}],$$

and thereby define an I -trajectory (E). The number of different symbols E_i is finite. We add the convention that a null set E_i is to be dropped from the trajectory (E). Derivation by generalized substitution includes all of the preceding types of derivation. We shall illustrate this fact for the case of derivation by association. Derivation by association is obtained if we set

$$\phi(x_1, \dots, x_m) = (x_1, \dots, x_m),$$

and understand that

$$(r_1, \dots, r_m) = (0, 1, \dots, m-1).$$

The symbol E_i reduces to the block

$$B_i = c_i c_{i+1} \dots c_{i+m-1},$$

as in the case of derivation by association.

We shall illustrate these processes further by deriving two recurrent trajectories from the trajectory T of the preceding section. We shall use the method of association letting $m = 2$. The generating symbols are the 2-blocks of T , namely 11, 12, 22, 21. We shall denote these blocks respectively by 1, 2, 3, 4 and write a few terms of the resulting trajectory. The symbols with indices $i \geq 0$ written above the corresponding symbols of T begin as follows:

$$\begin{array}{cccccccccccc} 2 & 3 & 4 & 2 & 4 & 1 & 2 & 3 & 4 & 1 & 2 & 4 & 2 & 3 & 4 & 2 & 4 & 1 & 2 \\ 1 & 2 & 2 & 1 & 2 & 1 & 1 & 2 & 2 & 1 & 1 & 2 & 2 & 1 & 2 & 2 & 1 & 2 & 1 \end{array}$$

Upon reducing the first of these trajectories mod 2 we obtain a trajectory which contains

$$0100010101000100010.$$

This trajectory is distinct from T since it has sequences of three equal symbols. These new trajectories are recurrent as we shall see.

A trajectory T' derived from a recurrent trajectory T by projection or reduction is clearly recurrent. The trajectory T' may however be periodic when T is non-periodic. The facts for the case of derivation by association are otherwise, as the following theorem shows.

THEOREM 9.1. *If (c') is a recurrent I -trajectory derived from a recurrent I -trajectory (c) by association in r -blocks, and $R'(n)$ and $R(n)$ are the recurrency functions of (c') and (c) respectively, then*

$$(9.1) \quad R'(n) = R(n + r - 1) - r + 1.$$

Moreover (c') is periodic if and only if (c) is periodic.

A block $[k + 1, k + n]$ of (c') is determined by the $(n + r - 1)$ -block $[k + 1, k + n + r - 1]$ of (c) and conversely. We term these blocks *corresponding*. Each $(n + r - 1)$ -block of (c) appears in each $R(n + r - 1)$ -block H of (c) . But H corresponds in the above sense to a block (c') of length

$$(9.2) \quad R(n + r - 1) - (r - 1),$$

while an $(n + r - 1)$ -block of (c) corresponds to an n -block of (c') . Hence $R'(n)$ is at most the integer (9.2).

To show that $R'(n)$ is at least the integer (9.1), let H be a minimax $(n + r - 1)$ -covering in (c) , that is an $R(n + r - 1)$ -block of (c) whose final $(n + r - 1)$ -block appears just once in H . Let K be the block of (c') corresponding to H . The final n -block of K corresponds to the final $(n + r - 1)$ -block of H and appears just once in K , so that $R'(n)$ is at least the integer (9.1), and the proof of (9.1) is complete.

If (c) is non-periodic, $R(n) > 2n$ in accordance with Theorem 7.5. It follows then from (9.1) that

$$R'(n) > 2n + r - 1.$$

The I -trajectory (c') is accordingly non-periodic, and the proof of the theorem is complete.

The preceding theorem leads to the following extension of Theorem 7.5.

THEOREM 9.2. *If $P(r)$ is the number of different r -blocks in a recurrent non-periodic I -trajectory (c) , then for $m \geq r$,*

$$(9.3) \quad R(m) \geq 2m + P(r) - r.$$

Let (c') be derived from (c) by association in r -blocks. If $R'(n)$ is the recurrency function of (c') , it follows from Theorem 7.5 that

$$R'(n) \geq 2n + P(r) - 1.$$

But from the preceding theorem we then see that

$$R(n + r - 1) - r + 1 = R'(n) \geq 2n + P(r) - 1,$$

from which (9.3) follows upon setting $m = n + r - 1$.

Before coming to the next theorem we shall need the following definition.

A definition. Let there be given a finite set S of q -blocks $B_1, \dots, B_p$ of a set of generating symbols. Let n be a positive integer less than q . A least integer $r(n) \leq q$ (if such exists) such that each n -block appearing in a block of the set S appears in each $r(n)$ -subblock of each of the blocks B_j will be called the n -th *recurrency index* of the set S . For example, let the blocks B_i be the blocks

$$(9.4) \quad B_1 = 1221, \quad B_2 = 2112, \quad B_3 = 1212, \quad B_4 = 2121.$$

These blocks have a recurrency index $r(1) = 3$. The recurrency index $r(2)$ does not exist.

The following theorem concerns the I -trajectory (c^*) derived from the I -trajectory (c) by substitution. The I -trajectory (c^*) is formed as described above by substituting the respective s -blocks A_i for the generating symbols a_i appearing in (c) . These blocks A_i are then expanded to form the I -trajectory (c^*) of the symbols a_i . We shall apply the preceding definition of the recurrency index of a set of blocks to the set S of $2s$ -blocks $A_i A_j$. Each m -block of (c^*) for which $m \leq s + 1$ lies in one of these $2s$ -blocks of (c^*) so that in the case where the set S has a recurrency index $r(m) \leq s + 1$ it is clear that the recurrency index $R^*(m)$ of (c^*) exists and is at most $r(m)$. This is affirmed in part (a) of the theorem. Part (b) conditions $R^*(n)$ on the lower side instead of on the upper side. This theorem is useful in the following section.

THEOREM 9.3. *Let (c^*) be an I -trajectory derived from an I -trajectory (c) by substitution as described above.*

(a) *If the set of $2s$ -blocks $A_i A_j$ of (c^*) has an m -th recurrency index $r(m) \leq s + 1$, then*

$$(9.5) \quad R^*(m) \leq r(m).$$

(b) *If some block A_i appears r times consecutively in (c^*) with $r > 1$, and if (c^*) does not have the period s , then*

$$(9.6) \quad R^*(s + 1) > sr,$$

or else $R^(s + 1)$ fails to exist.*

Statement (a) has already been proved. We therefore turn to the proof of (b).

It is assumed that the block $B = A_i \cdots A_i$, of length sr , appears in (c^*) . Since the length of A_i is s , any $(s + 1)$ -block of B has equal terminal elements.

If $R^*(s+1)$ exists and $R^*(s+1) \leq sr$, every $(s+1)$ -block of (c^*) has a copy in B . Thus every $(s+1)$ -block of (c^*) has equal terminal elements and (c^*) has the period s , contrary to hypothesis.

The proof of the theorem is complete.

The following theorem has a special interest in view of the mode of formation of the Morse recurrent trajectory. The trajectory H thereby affirmed to exist is not necessarily admissible. It will be admissible if the only symbols employed are α and β .

THEOREM 9.4. *Corresponding to an arbitrary recurrent ray*

$$R = a_0 a_1 a_2 \cdots,$$

there exists a recurrent trajectory H with R as subray and with the recurrency indices of R . The trajectory H is in general not uniquely determined by R .

Let B_r be the initial r -block of R , $r = 1, 2, \cdots$. The block B_r appears later in R immediately preceded by some r -block A_r . Of the blocks $A_1, A_2, \cdots$ there exists an infinite subset S_1 with last symbol in common. We take a_{-1} as this last symbol. Of the blocks of S_1 there exists an infinite subset S_2 with the second from the last symbol in common. We take a_{-2} as this common symbol. Proceeding in this way we define sets $S_3, S_4, \cdots$ and symbols $a_{-3}, a_{-4}, \cdots$ respectively. That the resulting trajectory H is recurrent and has the same recurrency indices as R follows from the fact that each block of H appears in R .

That H is not uniquely determined is shown by the trajectories T and T' of § 8 which have the recurrent ray R of (8.4) as a common subray.

10. Deferred recurrence. In this section we shall exhibit a recurrent trajectory E for which $R(n)$ exceeds a prescribed function $f(n)$ for infinitely many values of n but which is such that the limit inferior of $R(n)/n$ is finite. That a recurrent trajectory exists for which $R(n)$ exceeds a prescribed function $f(n)$ for all values of n has already been shown by Robbins ([1], Theorem 2). Our example shows the possibility of extreme variability in the indices $R(n)$. The trajectories E here defined form an extended class of which the trajectory T of § 8 is a member.

The generating symbols shall be the free symbols α and β . Let $r_0, r_1, \cdots$ be an arbitrary set of positive integers. We shall make use of powers of blocks, understanding that these powers are to be formally expanded. Set

$$\begin{array}{ll} A_0 = \alpha, & B_0 = \beta, \\ A_1 = A_0^{r_0} B_0^{r_0}, & B_1 = B_0^{r_0} A_0^{r_0}, \\ \cdot & \cdot \\ A_{n+1} = A_n^{r_n} B_n^{r_n}, & B_{n+1} = B_n^{r_n} A_n^{r_n}. \end{array}$$

Let s_n be the length of A_n and B_n . Observe that

$$(10.1) \quad s_{n+1} = 2s_n r_n.$$

Let (c) be an I -trajectory such that the initial s_n -block of the ray $c_0 c_1 c_2 \cdots$ is A_n for each n . This definition is self consistent since A_{n-1} is an initial block of A_n . We further set

$$(10.2) \quad c_{-m-1} = c_m \quad (m = 0, 1, \cdots).$$

Proceeding inductively we see that

$$\begin{aligned} A_n^{-1} &= A_n, & B_n^{-1} &= B_n, & (n \text{ even}) \\ A_n^{-1} &= B_n, & B_n^{-1} &= A_n, & (n \text{ odd}). \end{aligned}$$

Hence the s_n -block with final index -1 is A_n when n is even and B_n when n is odd. If the symbols of (c) are grouped into s_r -blocks with first indices $\equiv 0 \pmod{s_r}$, one obtains an I -trajectory E_r with generating symbols A_r and B_r .

If the integers r_i are all 1 and $\alpha = 1$, $\beta = 2$, (c) yields the recurrent trajectory T of § 8. We shall make use of the recurrency function $\rho(n)$ of T . The trajectory defined by (c) will be denoted by E . It is uniquely determined by the symbols α and β and the integers r_i .

THEOREM 10.1. *The trajectory E is recurrent and non-periodic with a recurrency function $R^*(n)$ such that*

$$(10.3) \quad s_{n+1} \leq R^*(s_n + 1) \leq 2s_{n+3}.$$

Let B be an arbitrary $(s_n + 1)$ -block of E . The block B appears in some expanded 2-block of E_n . But each such 2-block of E_n appears in both A_{n+3} and B_{n+3} as is easily seen. The latter two blocks are s_{n+3} -blocks of E and one of them at least is in each $2s_{n+3}$ -block of E . Hence the right-hand condition in (10.3) is valid. It follows that E is recurrent.

The left condition in (10.3) follows from (b) in Theorem 9.3. For the block A_n appears $2r_n$ consecutive times in E . Moreover E does not have the period s_n since E contains the block $A_n B_n$ and B_n is different from A_n . From (9.6) we then find that

$$R^*(s_n + 1) \geq 2r_n s_n = s_{n+1},$$

and the proof of (10.3) is complete. Moreover we see that

$$R^*(s_n + 1) \geq 2s_n$$

and it follows from Theorem 7.6 that E cannot be periodic.

COROLLARY. If $f(n)$ is an arbitrary positive integral function, it is possible to choose the r_n 's or any subset of the r_n 's so that for corresponding values of n ,

$$(10.4) \quad R^*(s_n + 1) > f(s_n + 1).$$

Recall that $s_{n+1} = 2s_n r_n$. In order that (10.4) may hold we have merely to take r_n so large that

$$s_{n+1} > f(s_n + 1)$$

and (10.4) will follow from (10.3).

We continue with the following theorem.

THEOREM 10.2. If the sequence of integers r_i used in defining E contains arbitrarily long blocks of 1's,

$$(10.5) \quad \liminf_{n \rightarrow \infty} \frac{R^*(n)}{n} \leq \frac{11}{2}.$$

If $\rho(m)$ is the recurrence function of the trajectory T of § 8, there exists a sequence of integers m_i becoming infinite with i and such that

$$(10.6) \quad \liminf_{i \rightarrow \infty} \frac{\rho(m_i)}{m_i} = \frac{11}{2}.$$

We suppose integers m_i so chosen. The sequence of integers r_i contains arbitrarily long blocks of 1's. It follows that there exists a sequence of positive integers

$$(10.7) \quad h_1 < k_1 < h_2 < k_2 < \dots,$$

such that for each i ,

$$(10.8) \quad r_{h_i} = r_{h_i+1} = \dots = r_{k_i} = 1.$$

Moreover these integers can be chosen so that

$$\mu_i = k_i - h_i > \rho(m_i).$$

We shall prove the following statement.

(α). The m_i -th recurrence index of E_{h_i} is at most the recurrence index $\rho(m_i)$.

The trajectory E_{h_i} is generated from the blocks A_{h_i}, B_{h_i} . These blocks combine to form the blocks A_{k_i}, B_{k_i} in exactly the way in which the symbols 1 and 2 combine to form the blocks a_{μ_i} and b_{μ_i} of the Morse trajectory T . In E_{h_i} the blocks

$$(10.9) \quad A_{k_i} B_{k_i}, \quad A_{k_i} A_{h_i}, \quad B_{k_i} A_{h_i}, \quad B_{k_i} B_{k_i},$$

regarded as blocks of the symbols A_{h_i} and B_{h_i} , accordingly have an m_i -th recurrency index $\rho(m_i)$. For

$$\rho(m_i) < k_i - h_i < 2^{k_i - h_i} = 2^{\mu_i},$$

and 2^{μ_i} is the length of the blocks A_{k_i} and B_{k_i} regarded as blocks of the symbols A_{h_i} , B_{h_i} . It follows from (a) of Theorem 9.3 that E_{h_i} has an m_i -th recurrency index at most $\rho(m_i)$, and the proof of (α) is complete.

We now regard E as derived from E_{h_i} by expansion of the blocks A_{h_i} and B_{h_i} of E_{h_i} into blocks of α 's and β 's. Set

$$(10.10) \quad n_i = s_{h_i}(m_i - 1).$$

An arbitrary n_i -block of E is found in some expanded m_i -block of E_{h_i} , and hence in each expanded $\rho(m_i)$ -block of E_{h_i} , and finally in each $[s_{h_i}\rho(m_i) + s_{h_i}]$ -block of E . Hence

$$(10.11) \quad R(n_i) \leq s_{h_i}[\rho(m_i) + 1].$$

It follows from (10.10), (10.11) and (10.6) that

$$\liminf_{i \rightarrow \infty} \frac{R(n_i)}{n_i} \leq \liminf_{i \rightarrow \infty} \frac{\rho(m_i) + 1}{m_i - 1} = \frac{11}{2},$$

and the proof of the theorem is complete.

The preceding theorems and corollary combine to give use the following.

THEOREM 10.3. *If $f(n)$ is an arbitrary positive integral function of n , then for suitable choices of the integers r_i defining E , the recurrency function $R^*(n)$ of E exceeds $f(n)$ for infinitely many values of n while*

$$\liminf_{n \rightarrow \infty} \frac{R^*(n)}{n} \leq \frac{11}{2}.$$

11. Transitive rays and their ergodic functions. We shall concern ourselves with rays of the form

$$c_0 c_1 c_2 \cdots$$

Such a ray will be termed transitive if it contains a copy of each admissible subblock. Transitive rays exist. If $A_1, A_2, \cdots$ is an enumeration of admissible blocks, the ray

$$X = A_1 e_1 A_2 e_2 \cdots$$

is admissible provided the symbols e_i are successively chosen in a suitable way. The ray X is clearly transitive. Let R be an arbitrary ray and let n be a positive integer. Suppose an initial r -block B of R contains each admissible n -block while no initial subblock of B has this property. The integer r will then be termed the n -th *transitivity index* $\phi(n)$ of R . If R is transitive, the n -th transitivity index of R exists for each value of n and will be termed the *ergodic function* $\phi(n)$ of R .

A block $H(n)$ of minimum length containing each admissible n -block will be termed a *minimum n -covering*. The length $\theta(n)$ of $H(n)$ will be termed the n -th *covering index*. In the case where the generating symbols are all free, Martin [1] has shown that for each n there exists a minimum n -covering in which no n -block is repeated. We shall presently extend Martin's result to the case where successive inverses are prohibited. We are concerned here with the existence of transitive rays whose ergodic function is in some sense relatively small. Noting that each ergodic function $\phi(n)$ satisfies the condition

$$\phi(n) \geq \theta(n),$$

we are led to the question, does there exist a transitive ray such that $\phi(n) = \theta(n)$ for each n ? The answer is no, in general, as we shall see.

We begin with the case where there are just two generating symbols, namely the free symbols α and β . In this case it follows from Martin's theorem that

$$\theta(n) = 2^n + n - 1.$$

The only minimum 1-coverings are $\alpha\beta$ and $\beta\alpha$. The only minimum 2-covering which starts with $\alpha\beta$ is $\alpha\beta\beta\alpha\alpha$. If we wish a minimum 3-covering which starts with this 5-block we may continue with β or α , but we must then continue with $\alpha\beta$ or $\beta\alpha\beta$ respectively, thereby obtaining the blocks

$$(\alpha\beta\beta\alpha\alpha)\beta\alpha\beta, \quad (\alpha\beta\beta\alpha\alpha)\alpha\beta\alpha\beta.$$

Further continuation subject to the condition that no 3-block repeats is impossible. Since $\theta(3) = 10$, there exists no transitive ray for which

$$\phi(n) = \theta(n) \quad (n = 1, 2, 3).$$

It is conceivable that there might exist a transitive ray for which $\phi(n) = \theta(n)$ for all values of n exceeding some fixed integer m . That this is impossible is shown by the following previously unpublished theorem of R. Oldenburger.

THEOREM 11.1. *If the generating symbols reduce to two free symbols α and β there exists no transitive ray whose ergodic function $\phi(n)$ equals the covering index $\theta(n)$ for two successive values of $n > 1$.*

The proof as given by Oldenburger is essentially as follows.

We assume the theorem false. There then exists a minimum n -covering $H(n)$, $n > 2$, which starts with a minimum $(n-1)$ -covering $H(n-1)$. No n -block repeats in $H(n)$, nor $(n-1)$ -block in $H(n-1)$. We shall arrive at a contradiction.

Since α and β can be interchanged throughout it is no essential restriction to assume that in $H(n-1)$ the $(n-1)$ -block β^{n-1} occurs before α^{n-1} . We shall then show that $H(n-1)$ has the form

$$(11.1) \quad H(n-1) = [\beta^{n-1}\alpha \cdot \cdot \cdot \beta\alpha^{n-1}],$$

and that $H(n)$ has the form

$$(11.2) \quad H(n) = H(n-1)[\alpha\beta \cdot \cdot \cdot \alpha\beta^n].$$

The block of $H(n)$ which remains when $H(n-1)$ is removed will be denoted by J . To establish the correctness of the representations (11.1) and (11.2) one reasons as follows.

The block α^n appears in $H(n)$. It cannot appear wholly in $H(n-1)$, for this would imply repetition of α^{n-1} in $H(n-1)$, and thus $\beta\alpha^n$ appears in $H(n)$. Since β^{n-1} precedes α^{n-1} in $H(n-1)$, $\beta\alpha^{n-1}$ appears in $H(n-1)$. Since $\beta\alpha^{n-1}$ appears just once in $H(n)$, it must appear as the initial n -block of $\beta\alpha^n$, and since this latter block is not wholly in $H(n-1)$, $H(n-1)$ ends with $\beta\alpha^{n-1}$ and J begins with α . This α is of course followed by β .

The block β^n is in $H(n)$ and not wholly in $H(n-1)$. Since J begins with α , β^n must appear wholly in J , and hence $\alpha\beta^{n-1}$ and $\alpha\beta^n$ appear in J . It follows that β^{n-1} , which appears in $H(n-1)$, must appear at the left of $H(n-1)$; otherwise it would be preceded by α , and $\alpha\beta^{n-1}$ would appear twice in $H(n)$. Hence $H(n-1)$ begins with $\beta^{n-1}\alpha$. If J does not end with β^n , $\beta^n\alpha$ appears in J , and $\beta^{n-1}\alpha$ appears both in J and $H(n-1)$. Hence J ends with $\alpha\beta^n$ and the representations (11.1) and (11.2) are correct.

We now make use of the assumption that $n > 2$. Then $n-2 > 0$ and $\alpha\beta^{n-2}$ occurs in $H(n-1)$. This block cannot be followed by β without repetition of β^{n-1} in $H(n-1)$. Hence $\alpha\beta^{n-2}\alpha$ appears in $H(n-1)$. But $\beta^{n-2}\alpha$ already appears at the beginning of $H(n-1)$. Hence $H(n)$ cannot start with a minimum $(n-1)$ -covering, and the proof of the theorem is complete.

We turn to the case where the only condition on the generating symbols is the restriction of inverses. This is the simplest case of strict dynamical significance. It arises in the case of surfaces of negative curvature with boundaries. Cf. Morse [1]. In this case one has a set of generating symbols of the form

$$(11.3) \quad a_1, \dots, a_p; a_1^{-1}, \dots, a_p^{-1} \quad (p > 1),$$

with the restriction that successive symbols shall not be inverses. We shall prove the following theorem by suitably modifying Martin's proof.

THEOREM 11.2. *For generating symbols of the form (11.3), subject to no condition beyond the restriction of inverses, there exists a minimum r -covering in which no r -block repeats.*

The following rule for the formation of a minimum r -covering $H(r)$ was first given by Martin for the case of free generating symbols $c_1, \dots, c_n$. Start with c_1^{r-1} and continue successively adjoining the generating symbol of highest index consistent with the condition that no r -block be repeated. This will lead to a minimum r -covering in the case of free symbols c_i . For the case of inverses we adjoin the condition that successive symbols shall not be inverses. With this condition added the process will not always lead to a minimum r -covering. We are able however to prove the following lemma.

LEMMA 11.1. *A necessary and sufficient condition that the application of the Martin rule, subject to the restriction of inverses, shall lead to a minimum r -covering is that c_1 and c_2 shall not be inverses in the set $c_1, \dots, c_n$ of generating symbols.*

Let M_r be the block of maximum length obtained by applying the Martin rule subject to the restriction of inverses. If A is any admissible $(r-1)$ -block, the r -blocks

$$(11.4) \quad Ac_n, \dots, Ac_1$$

appear in M_r , if at all, in the order written. One of the blocks (11.4) is inadmissible since the last symbol in A cannot be followed by its inverse. If Ac_1 is in M_r , all other admissible blocks in (11.4) are in M_r . If Ac_1 is not admissible, Ac_2 is admissible and is in M_r only if all of its predecessors in (11.4) are in M_r . We set $E = c_1^{r-1}$. We continue with a proof of (a) and (b).

(a). *The block M_r ends with E .* If M_r ended with an $(r-1)$ -block $A \not\equiv E$, M_r could not begin with A , since M_r begins with E . A would then

appear in M_r at most $n - 1$ times; otherwise some one of the $n - 1$ admissible blocks $c_i A$ would appear twice. Following A 's last appearance we can continue M_r with c_2 (or c_1 if $A c_2$ is inadmissible), contrary to the hypothesis that M_r ends with A . Hence M_r ends with E .

(b). *The block M_r ends with Ec_1 .* Suppose M_r ends with an r -block $B \neq Ec_1$. Set $B = bE$ in accordance with (a). We are supposing that $b \neq c_1$. The block E appears n times in M_r ; otherwise M_r could be continued. Hence Ec_1 appears in M_r but not at the end. But Ec_1 cannot be continued in accordance with the Martin rule. From this contradiction we infer the truth of (b).

We shall now make use of the assumption that c_1 and c_2 are not inverses and show that M_r contains all r -blocks.

We first observe that each admissible block Ec_j appears in M_r since M_r ends with Ec_1 . Hence M_r contains n copies of E . It follows that M_r contains each admissible block $c_j E$. Let $B = b_1 \cdots b_r$ be an arbitrary admissible r -block. We shall show that M_r contains B .

Suppose that M_r does not contain B . Set $b_2 \cdots b_r = D$. Then $D \neq E$ by virtue of (b). Hence D appears at most $n - 2$ times in M_r ; otherwise each admissible block $c_j D$ would appear in M_r , including B . Hence Dc_1 does not appear in M_r (or Dc_2 , if Dc_1 is inadmissible). We now apply the same reasoning to Dc_1 (or Dc_2) that we have applied to B and infer that $b_3 \cdots b_r c_1 c_1$ (or $b_3 \cdots b_r c_2 c_1$) does not appear in M_r . Continuing we arrive at the conclusion that c_1^r does not appear in M_r , contrary to (b).

The condition of the lemma is accordingly sufficient.

To prove the condition necessary we suppose that $c_1 = c_2^{-1}$. If M_r were an r -covering c_2^r would appear in M_r , but not in the last position, by virtue of (a). But c_2^r cannot be continued without violating the Martin rule, or else the condition on inverses. Hence the condition is necessary, and the proof of the lemma is complete.

Theorem 11.2 is an immediate consequence of the lemma.

If there are n generating symbols subject merely to the restriction of inverses, there are $n(n - 1)^{r-1}$ different admissible r -blocks so that the r -th covering index $\theta(n)$ has the value

$$\theta(r) = n(n - 1)^{r-1} + r - 1.$$

With this understood we shall prove the following theorem.

THEOREM 11.3. *When there are n generating symbols subject merely to*

the restriction of inverses, each ergodic function $\psi(r)$ is at least $\theta(r)$ and there exists an ergodic function $\phi(r)$ such that

$$(11.5) \quad \phi(r) < \left(\frac{n-1}{n-2} \right) n(n-1)^{r-1} < \frac{n-1}{n-2} \theta(r).$$

Let H_r be a minimum r -covering and let H^*_r be the block obtained from H_r by omitting the first $r-1$ symbols c_1 . The ray

$$R = H_1 H^*_2 H^*_3 \cdots$$

is admissible and transitive. We note in particular that H^*_r is preceded by c_1^{r-1} in R . For this ray $\phi(r)$ is at most the length of the block $H_1 H^*_2 \cdots H^*_r$, that is

$$\phi(r) \leq \sum_{m=1}^r n(n-1)^{m-1}.$$

Hence

$$\phi(r) \leq n \left[\frac{1 - (n-1)^r}{1 - (n-1)} \right] = \left(\frac{n-1}{n-2} \right) n \left[(n-1)^{r-1} - \frac{1}{n-1} \right],$$

and (11.5) follows at once.

We conclude this section by proving the following theorem.

THEOREM 11.4. *If $N(r)$ is the number of admissible r -blocks under our general conditions of admissibility, there exists an ergodic function $\phi(r)$ such that*

$$(11.6) \quad \phi(r) < (r+1)N(r),$$

while each ergodic function is at least $N(r) + r - 1$.

We begin by proving the following.

(a). *There exists an r -covering of length less than $(r+1)N(r)$.*

Statement (a) is true if $r=1$. For if $a_1, \dots, a_n$ is the set of generating symbols, there exists a 1-covering of the form $a_1 e_1 a_2 e_2 \cdots e_{n-1} a_n$ provided the symbols e_i are suitably chosen. This 1-covering has the length $2n-1 < 2N(1)$. We assume then that A_r is an r -covering of length less than $(r+1)N(r)$. We shall show how to form an $(r+1)$ -covering of length less than $(r+2)N(r+1)$.

There are $N(r)$ distinct r -blocks in A_r and hence at least $N(r) - 1$ distinct $(r+1)$ -blocks. There remain at most

$$\mu = N(r+1) - N(r) + 1$$

$(r+1)$ -blocks, say $B_1, \dots, B_\mu$ not in A . The block

$$A_{r+1} = A_r e_1 B_1 e_2 B_2 e_3 \dots e_\mu B_\mu,$$

in which the symbols e_i are chosen so as to make A_{r+1} admissible, is an $(r+1)$ -covering. Let s_i be the length of A_i . We see that

$$s_{r+1} = s_r + (r+2)\mu.$$

By virtue of the hypothesis that $s_r < (r+1)N(r)$ we infer that

$$\begin{aligned} (11.7) \quad s_{r+1} &< (r+1)N(r) + (r+2)[N(r+1) - N(r) + 1] \\ &= (r+2)N(r+1) - N(r) + r + 2. \end{aligned}$$

But for $r > 1$,

$$N(r) \geq 2^r \geq r + 2,$$

so that (11.7) yields the relation

$$s_{r+1} < (r+2)N(r+1),$$

and the proof of (a) is complete.

In proving (a) we have defined r -coverings A_r , $r = 1, 2, \dots$, such that A_r is an initial block of A_{r+1} . A ray whose first s_r symbols are those of A_r for each r will satisfy (11.6). The final statement of the theorem is immediate.

12. Category and sets of transitive rays. In this section we shall determine the category of the set of transitive rays or of sets of relatively transitive rays. More particularly, we shall determine the category of various subsets of transitive rays defined by limitations on the ergodic functions. In this way we obtain an idea of the distribution of transitive rays. A similar use can be made of measure theory (cf. § 13), but the development of this phase of the theory is deferred to a later paper.

In the category theory, the space used will be the space S of admissible rays with the Baire metric. Cf. § 4. Recall that the space S is compact, perfect, and totally disconnected. Let H be a perfect subset of S . A set A is of the *first category* relative to H if it is the sum of an at most enumerable set of nowhere dense (relative to H) subsets of H . These subsets will be called the components of A . The components of A may vary from the extreme

case where they are all null to the case where they are perfect. The complement $H - A$ of a set A of the first category is termed a *residual set* of the second category. Such a set is everywhere dense in H and has the power of the continuum. Cf. Hobson [1], p. 132.

Let R be a given ray. The rays obtained by deleting a finite initial block of R will be said to be *based* on R , those obtained by the inverse process of adding a finite block to R will be termed *extensions* of R . The rays which are limit rays in the space S of the set of rays based on R will be called *limit rays* of R . A ray R which is a limit ray of itself will be said to be *transitive relative to its limit rays*. It is thereby necessary and sufficient that R be non-periodic and that every block of R appear infinitely many times in R . The limit rays of a relatively transitive ray R form a closed set H . This set is identical with its derived set since each ray based on R is a limit ray of the set. The set H is accordingly perfect. Concerning the set H we have the following theorem.

THEOREM 12.1. *In the set H of limit rays of a relatively transitive ray R , the rays which are transitive relative to H form a residual set in H .*

Let $A_1, A_2, \dots$ be an enumeration of the blocks of R . Let E_n be the subset of rays of H which contain no copy of A_n . The set E_n is closed and contains no rays based on R . It is nowhere dense in H since the rays based on R are everywhere dense in H . Finally the set S of rays which are transitive relative to H is of the form

$$S = H - \sum E_n,$$

and the theorem follows from the definition of a residual set in H .

The above sets E_n may all be null, in fact are all null if R is a subray of a non-periodic recurrent trajectory. In the case where R is transitive, that is transitive relative to the set of all admissible rays, the sets E_n are never null as the following lemma shows.

LEMMA 12.1. *The set K of rays which exclude an admissible r -block B for which $r > 2$ is closed, nowhere dense, and of the power of the continuum.*

It is clear that K is closed. That K is nowhere dense in S follows from the existence of a transitive ray R . For the rays based on R are everywhere dense in S and transitive. Since K is closed, it must then be nowhere dense.

It remains to prove that K is of the power of the continuum. To that end we consider the admissible rays

$$(12.1) \quad \alpha^{r_1} \beta \alpha^{r_2} \beta \alpha^{r_3} \beta \dots, \quad r_i = 2 \text{ or } 3,$$

formed from the free symbols α and β . This set is in one-to-one correspondence with the dyadic decimals

$$.r_1r_2r_3\cdots, \quad r_i = 2 \text{ or } 3,$$

and hence is of the power of the continuum.

Similarly, the set

$$(12.2) \quad \beta^{r_1}\alpha\beta^{r_2}\alpha\beta^{r_3}\cdots, \quad r_i = 2 \text{ or } 3,$$

consists of admissible rays and is of the power of the continuum.

A ray of the form (12.1) has no 3-blocks in common with a ray of the form (12.2). Since the length of B is at least 3, it cannot appear in a ray of (12.1) and a ray of (12.2). Hence at least one of the sets (12.1) and (12.2) is contained in K . Thus K must be of the power of the continuum.

The set of non-transitive rays is of the first category. But much more than this can be said, as the following theorem shows.

THEOREM 12.2. *The set Σ of non-transitive rays is everywhere dense in S and is the sum of countably many nowhere dense, closed sets, each of which is of the power of the continuum.*

That the set Σ is everywhere dense follows from the fact that an arbitrary block B and a sequence A of the symbols α can be combined in the form BeA to form a non-transitive ray. Rays of this form are everywhere dense in S . Let $A_1, A_2, \cdots$ be an enumeration of admissible r -blocks with $r > 2$. The set E_n of rays which exclude A_n is closed, nowhere dense and of the power of the continuum, in accordance with Lemma 12.1. The set Σ is the sum of the sets E_n , and the proof is complete.

THEOREM 12.3. *If $f(n)$ is an arbitrary positive function of the positive integers n , the set F of transitive rays with ergodic function $\phi(n) > f(n)$, for all values of n , has the power of the continuum.*

Let $A_1, A_2, \cdots$ be an enumeration of admissible blocks consistent with their lengths. Let $r_1, r_2, \cdots$ be an arbitrary sequence of positive integers and let $c_1c_2\cdots$ be an arbitrary sequence of the symbols α and β . We introduce the block

$$B_i = \alpha^{r_1}c_i\alpha^{r_1}e_iA_i \quad (i = 1, 2, \cdots),$$

choosing e_i so that B_i is admissible. We then introduce the ray

$$E = B_1 b_1 B_2 b_2 \cdots,$$

choosing the symbols b_i so that the ray E is admissible. It is clear that E is transitive.

Let $\phi(n)$ be the ergodic function of E . We choose r_1 so that $\phi(n) > f(n)$ for $n = 1, 2, 3$. Let B be the initial $f(4)$ -block of E . If r_2 is chosen sufficiently large, blocks $A_1, A_2, \cdots, A_p$ entering into the definition of B will all be 1-blocks. In fact, if r_2 is sufficiently large only A_1 will enter into the definition of B . If A_i is a 1-block, $e_i A_i b_i$ is a 3-block so that B cannot contain β^4 . With r_2 so chosen, $\phi(4) > f(4)$.

Proceeding inductively with $m > 2$ we choose r_m so large that in the initial $f(m+2)$ -block C of E , at most $(m-1)$ -blocks A_i are employed. The corresponding blocks $e_i A_i b_i$ are at most $(m+1)$ -blocks so that β^{m+2} does not appear in C . Hence

$$(12.3) \quad \phi(m+2) > f(m+2) \quad (m = 3, 4, \cdots).$$

Thus $\phi(n) > f(n)$ for all values of n .

The above symbols c_i can be chosen arbitrarily from the set α, β , the r_i 's remaining fixed. For fixed r_i 's there are then as many different rays E as there are different sequences $c_1 c_2 \cdots$, that is a number equal to the power of the continuum.

The proof of the theorem is complete.

The set F of the theorem is not closed. The set of transitive rays not in F is open in the set of all transitive rays so that F is everywhere dense only if it includes all transitive rays.

THEOREM 12.4. *Given an arbitrary function $f(n)$, the set ω of transitive rays R with ergodic functions $\phi(n) \leq f(n)$ for n exceeding some integer dependent on R is a set of the first category.*

Let D_m be the set of transitive rays for which $\phi(n) \leq f(n)$ for n exceeding m . The set D_m is closed. But we have seen in Theorem 12.2 that the set of intransitive rays is everywhere dense. Hence D_m is nowhere dense. But $\omega = \Sigma D_m$, and the proof is complete.

Let $H(n)$ be a block of minimum length $\theta(n)$ covering all admissible n -blocks. We have termed $\theta(n)$ the n -th covering index. As we have seen in § 11, there is in general no transitive ray with ergodic function $\phi(n) = \theta(n)$. We have however the following theorem.

THEOREM 12.5. *The set G of transitive rays with ergodic function $\phi(n)$ such that*

$$(12.4) \quad \liminf_{n \rightarrow \infty} \frac{\phi(n)}{\theta(n)} = 1,$$

where $\theta(n)$ is the n -th covering index, is a residual set of the second category.

We begin with a proof of the following.

(a). The set G is not empty.

To prove (a) let δ_n be a sequence of positive constants converging to zero. We shall consider a ray of the form

$$R = H(r_0)e_1H(r_1)e_2H(r_2) \cdots,$$

where $H(n)$ is defined as above and the symbols e_i are chosen so that R is admissible. The integers r_i are chosen as follows. Let $r_0 = 1$. Proceeding inductively we suppose that r_{n-1} is defined and that the block

$$H(r_0)e_1H(r_1)e_2 \cdots e_{n-1}H(r_{n-1})e_n$$

has the length m_n . We then choose r_n so large that

$$(12.5) \quad \frac{\theta(r_n) + m_n}{\theta(r_n)} < 1 + \delta_n.$$

If $\phi(n)$ is the ergodic function of the resulting ray R , it is clear that

$$\phi(r_n) \leq \theta(r_n) + m_n,$$

since the initial block of length $\theta(r_n) + m_n$ in R contains $H(r_n)$ and hence all r_n -blocks. The function $\phi(n)$ will satisfy (12.4) by virtue of (12.5), and the proof of (a) is complete.

If A is an arbitrary admissible r -block, the ray $X = AeR$ is admissible if e is suitably chosen. Moreover the ray R and the ray X have ergodic functions ϕ and ψ respectively such that

$$\theta(n) \leq \psi(n) \leq \phi(n) + r + 1.$$

Hence X satisfies the condition (12.4) and is in G . We conclude that the rays of G are everywhere dense.

To come to a proof of the theorem let m be a positive integer and p a rational number > 1 . The rays for which the transitivity index $\psi(m)$ satisfies the condition

$$\psi(m) > \theta(m)p$$

form a closed set D_m^p . The intersection

$$E_q^p = D_q^p D_{q+1}^p D_{q+2}^p \cdots$$

is closed. The complement of G is the sum of the sets E_q^p as q ranges over all positive integers and p ranges over all rational numbers > 1 . But G is everywhere dense so that E_q^p is nowhere dense, and the proof of the theorem is complete.

THEOREM 12. 6. *The set Z of transitive rays with ergodic functions $\phi(n)$ such that*

$$\liminf_{n \rightarrow \infty} \frac{\phi(n)}{\theta(n)} > 1$$

is everywhere dense, has the power of the continuum, and is of the first category.

The set Z includes the set of transitive rays for which $\phi(n) > 2\theta(n)$, and so has the power of the continuum in accordance with Theorem 12. 3. The set Z is a subset of the complement of the residual set G of Theorem 12. 5 and so is of the first category. It remains to show that Z is everywhere dense.

If R is a ray of Z , any ray R' based on R is again transitive. If the initial symbol of R' is the $(r+1)$ -st symbol of R , the ergodic functions $\psi(n)$ and $\phi(n)$ of R' and R , respectively, satisfy the conditions

$$\psi(n) \geq \phi(n) - r, \quad \liminf_{n \rightarrow \infty} \frac{\psi(n)}{\theta(n)} > 1.$$

Hence R' belongs to Z . But the rays R' based on the transitive ray R are everywhere dense so that Z is everywhere dense as stated.

The proof of the theorem is complete.

13. Trajectories with the separation property. The results of this section will form the basis of a more extensive treatment to be presented later. No proofs will be given in this place.

Let t be a trajectory involving two generating symbols α and β . By the α -length (β -length) of a block B of t we shall mean the number of α 's (β 's) in B . The trajectory t will be said to have the *separation property* and be of *type S* if the following holds.

The α -lengths of any two n -blocks of t differ by at most 1.

The preceding condition implies the corresponding condition for β -lengths. Trajectories of type S arise in many ways. In particular let

$$(13.1) \quad \frac{d^2y}{dx^2} + f(x)y = 0$$

be a differential equation in which $f(x)$ is continuous and has the period 1. Let $u(x)$ be a solution of (13.1) which vanishes at least once, but not identically. We are concerned with the distribution of the zeros of $u(x)$. Without loss of generality we can suppose that none of these zeros occur at the points $x = n$, where n is an integer, positive, negative or zero. For the zeros of $u(x)$ are countable, and for our purposes any transformation of the form $\bar{x} = x + c$ is admissible. With this understood let a trajectory t be defined as follows. Let each point on the x -axis at which $u(x) = 0$ be replaced by the symbol α , and each point $x = n$ be replaced by the symbol β . The resulting trajectory t will be of type S . With trivial exceptions every trajectory of type S can be obtained in this way. The study of trajectories of type S is thus a study of the zeros of solutions of equations of the form (13.1).

Let $\alpha_n(\beta_n)$ denote the α -length (β -length) of some n -block of a trajectory t of type S . It can be shown that there exists a number θ , $0 \leq \theta \leq 1$, such that

$$\lim_{n \rightarrow \infty} \frac{\alpha_n}{n} = \theta, \quad \lim_{n \rightarrow \infty} \frac{\beta_n}{n} = 1 - \theta.$$

The number θ will be termed the *rotation number* of t . Cf. Poincaré [1].

The trajectory is not uniquely determined by the rotation number. If θ is irrational, there are infinitely many trajectories of type S having θ as rotation number and these trajectories are all recurrent. Thus trajectories of type S are in general recurrent. A simple exceptional example is the trajectory in which all the symbols are α except for one β , in which case the rotation number is 1.

Given θ , $0 \leq \theta \leq 1$, the recurrent trajectories of type S which have θ as rotation number form a minimal set which consists of a single periodic trajectory if and only if θ is rational. Since all of the trajectories of one of these minimal sets have the same recurrency function, it can be denoted by $R_\theta(n)$. The behavior of $R_\theta(n)$ varies widely with the choice of θ , but in all cases

$$\liminf_{n \rightarrow \infty} \frac{R_\theta(n)}{n} \leq 4.$$

There exist irrational values of θ , and thus non-periodic recurrent trajectories such that

$$\limsup_{n \rightarrow \infty} \frac{R_\theta(n)}{n} \leq 6.5.$$

On the other hand, given an arbitrary function $f(n)$, there exist irrational values of θ such that $R_\theta(n) > f(n)$ for infinitely many values of n .

If certain exceptional θ -sets of (Lebesgue) measure zero are excluded, $R_\theta(n)$ displays great regularity in its asymptotic behavior. We can state that, except for a θ -set of measure zero,

$$\limsup_{n \rightarrow \infty} \frac{R_\theta(n)}{n} = +\infty,$$

while if $c > 1$, except for a θ -set of measure zero,

$$\limsup_{n \rightarrow \infty} \frac{R_\theta(n)}{n^c} \leq 1.$$

14. Typical open questions. Let R be a ray of the form $c_0 c_1 c_2 \cdots$ and let t be a trajectory containing R as a subray. Among the positive limit trajectories of t are found one or more distinct minimal sets of trajectories. Cf. Theorem 6.1. Suppose v_1 of these sets are periodic trajectories, and v_2 of these sets general minimal sets. The number v_1 is at most aleph null, and v_2 is at most aleph. That v_2 may equal aleph will be shown by an example in a later paper. The numbers v_1 and v_2 are uniquely determined by R and will be called the *limit indices* of R of the first and second types respectively. Our first question is as follows.

(1). What are the limit indices of algebraic numbers, regarding decimal representations as rays? Are these rays ever transitive or recurrent? What are the answers to similar questions concerning π and e ?

(2). Let t be a non-periodic recurrent trajectory. In all known examples $R(n)/n$ fails to converge. Is this true in general?

(3). If t is a non-periodic recurrent trajectory, $R(n) \geq kn$ where $k \geq 2$. Cf. Theorem 7.5. What is the least value of k such that $R(n) \geq kn$ for each non-periodic recurrent trajectory?

(4). In the case of general admissibility certain blocks and finite sequences of blocks are prohibited. Knowing these prohibitions, is it possible to give a constructive rule for the formation of a minimum block covering all admissible r -blocks, and what is the most general such r -covering?

(5). Does there exist an ergodic function $\phi(n)$ such that the ratio of $\phi(n)$ to $\theta(n)$ tends to the limit 1? Cf. Theorem 12.5.

(6). Is a theorem similar to that of Oldenburger true in case the generating symbols suffer the restriction of inverses, or in the general case?

(7). What is the most general minimal set with a given recurrency function $R(n)$? With permutation index $P(n)$? With $P(n)$ and $R(n)$ both given? These questions should be answered first when there are just two generating symbols.

These questions seem to point to an algebraic analysis. Questions more intimately connected with measure theory, or the analysis of space forms will be presented in a later paper.

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